



**Dat klopt voor jou!**

Cryptography: Everything you want to know, but are afraid to ask

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# Cryptography Use Case

1 Confidentiality

2 Integrity

3 Authentication

4 Access Control

4 Digital Signature

5 Digital Certificate

6 Non-repudiation

7 Blockchain



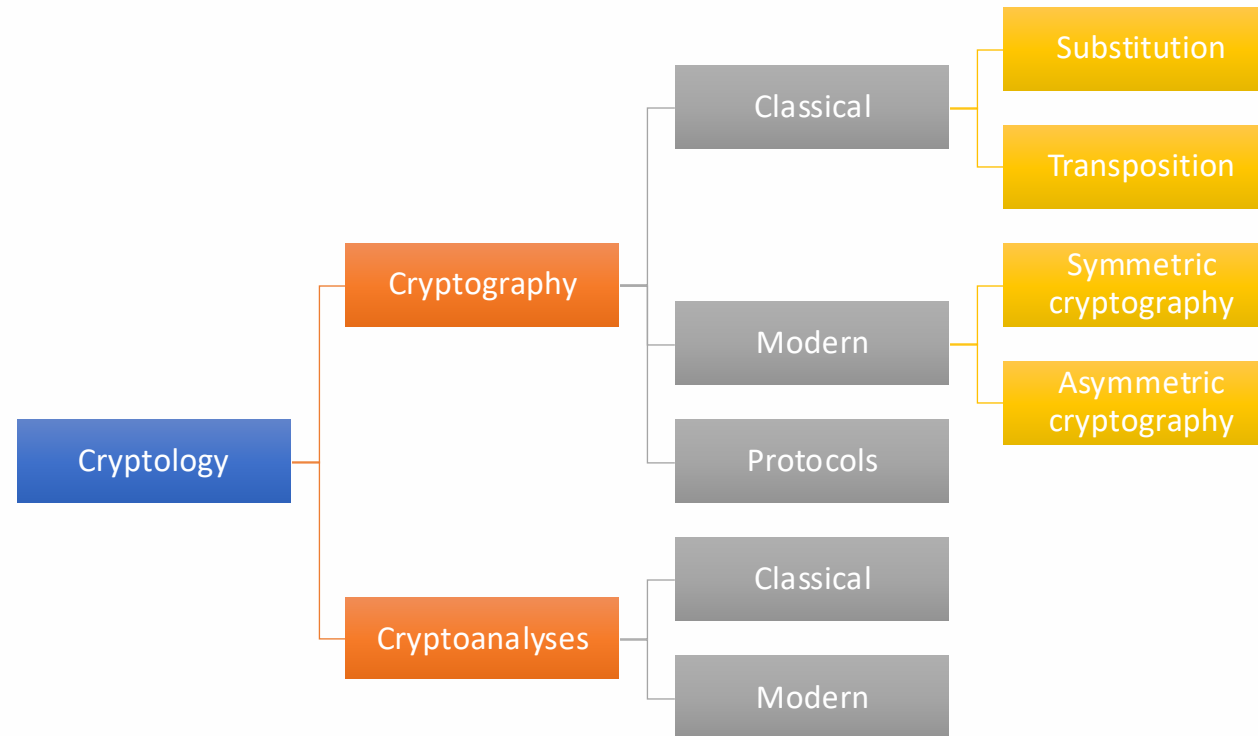
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# History of Cryptography

## Encryption



# The Study of Cryptology



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# Cryptographic Concepts

*Cryptography* is the science of using mathematics to encrypt and decrypt data

Cryptography enables you to store sensitive information or transmit it across insecure networks (like the Internet) so that it cannot be read by anyone except the intended recipient



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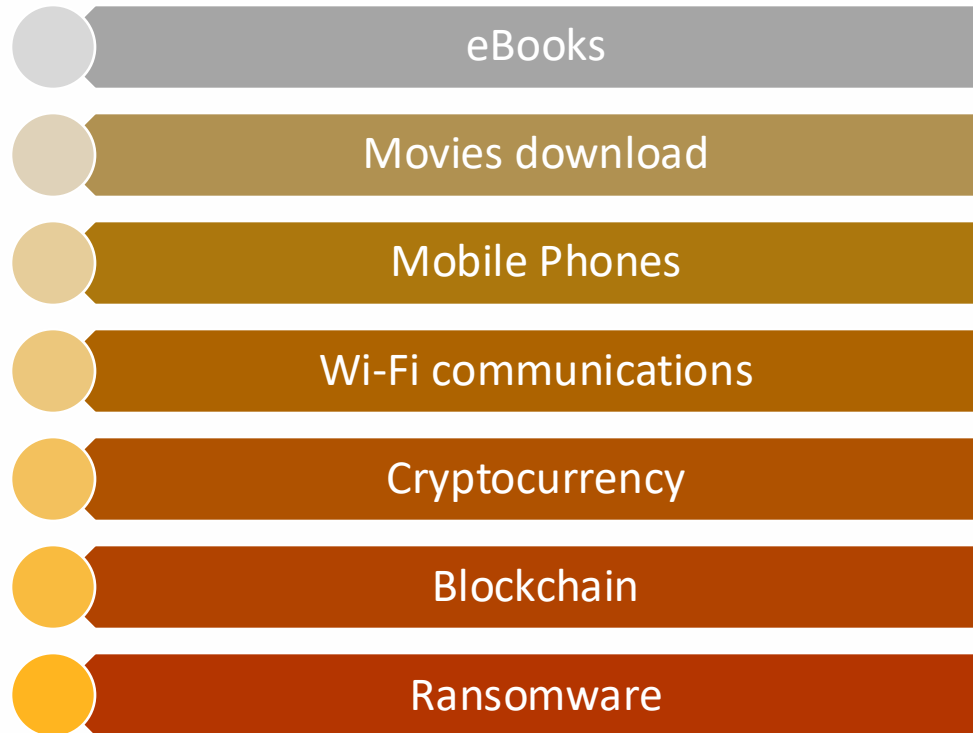
# Modern Use of Cryptography

|                 |                                                     |
|-----------------|-----------------------------------------------------|
| Confidentiality | • Encryption                                        |
| Data Integrity  | • Hash functions                                    |
| Authentication  | • Message authentication<br>• Entity authentication |
| Non-repudiation | • Digital signature                                 |
| Crypto systems  | • Block chaining                                    |



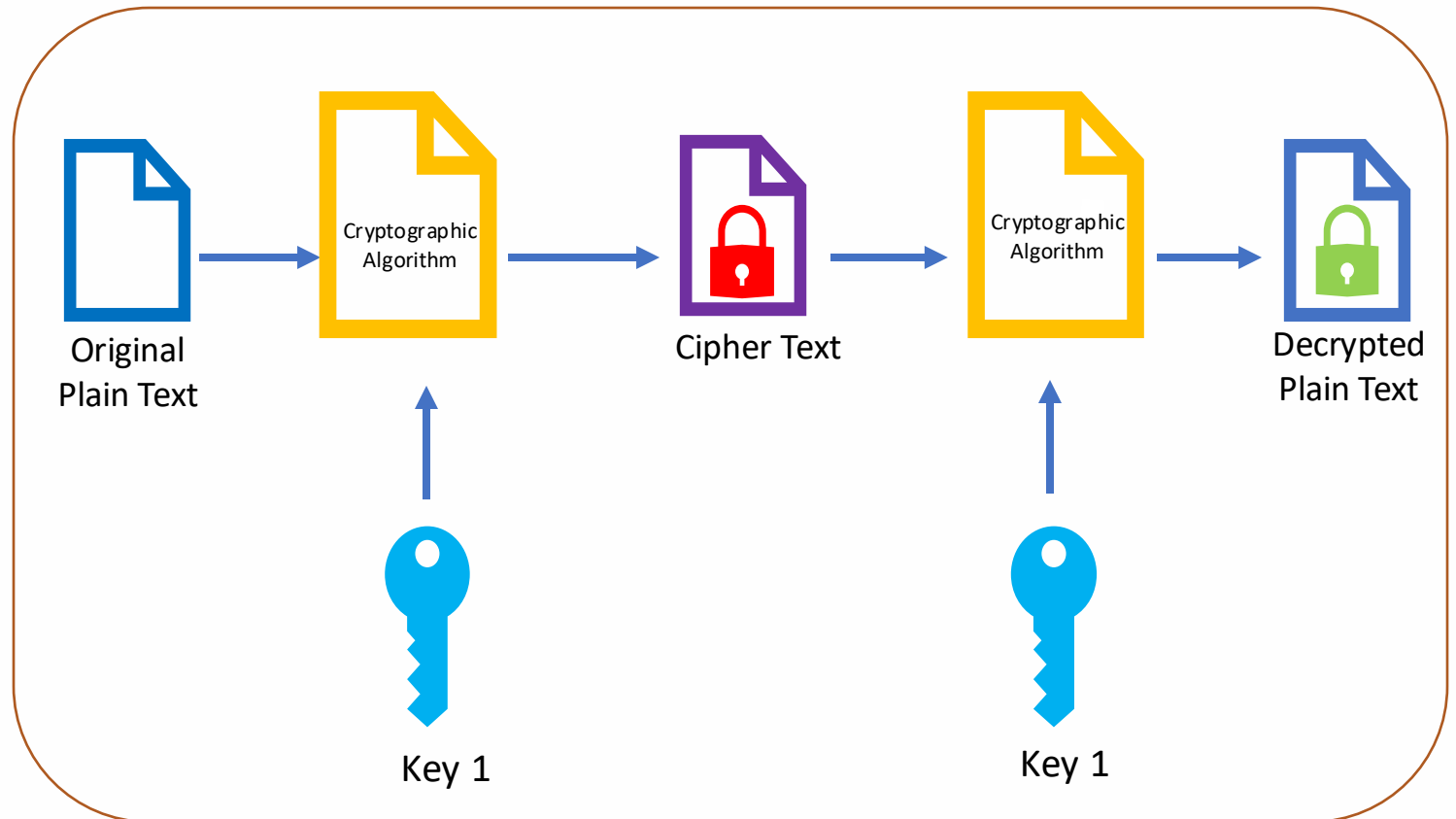
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# Products Using Modern Cryptography

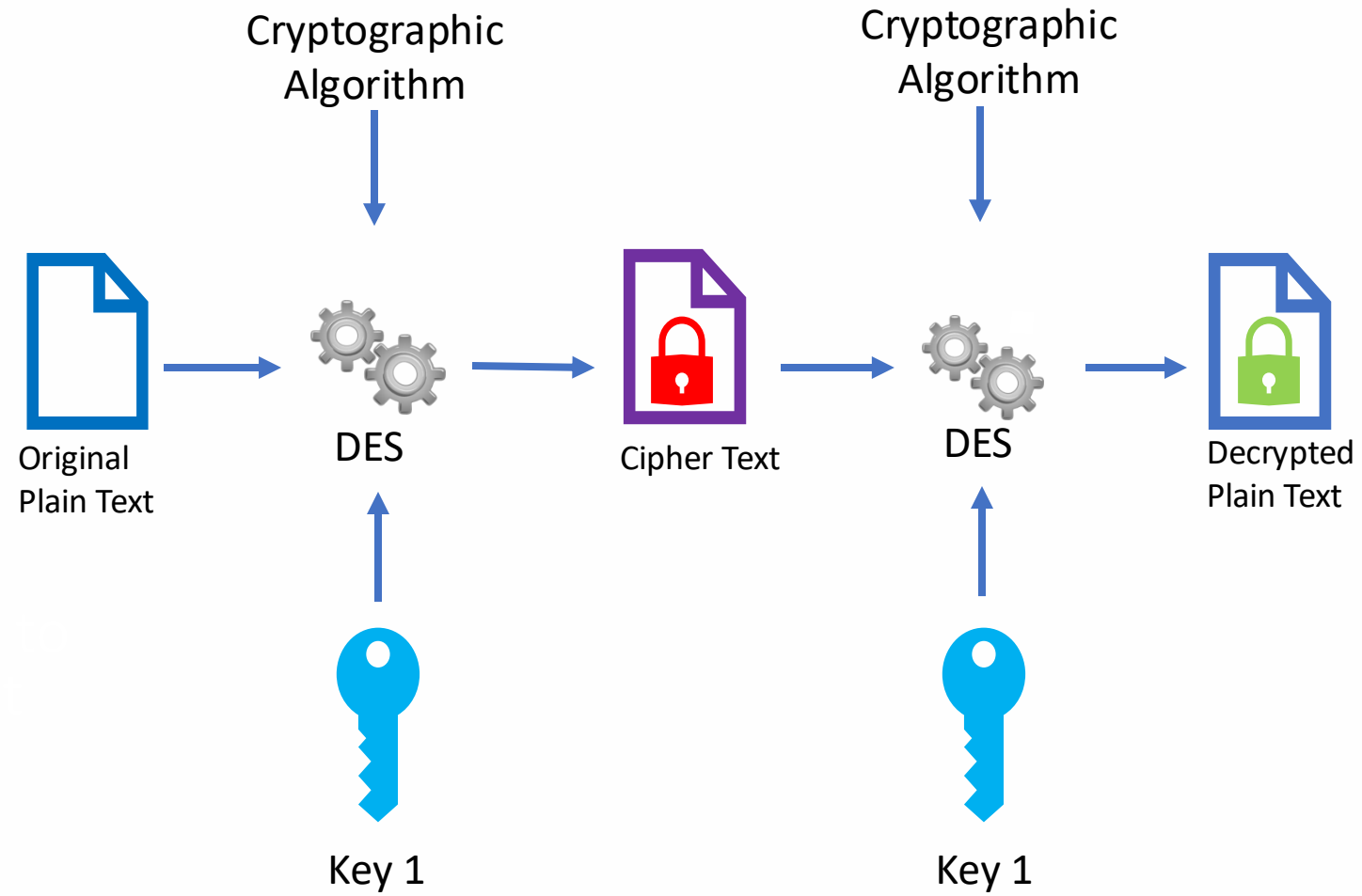


# Symmetric Key

Using the same key to encrypt and decrypt information also called secret key or symmetric key cryptography







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# The Challenge of Symmetric Cryptography

- ✓ Key distribution
- ✓ The same key [secret key] is used for encryption/decryption
- ✓ Both parties share the same key
- ✓ Each pair has to have a unique secret key
- ✓ Number of keys =  $n(n-1)/2$



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# Advantages of Symmetric Cryptography



Extremely fast, much faster than Asymmetric Cryptography



Can encrypt/decrypt very large amount of data



Hard to break if a large key size is used



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# Disadvantages of Symmetric Cryptography



Requires a secure out-of-band medium to share the keys



As the number of parties increases, manageability of keys is a problem



Provides only confidentiality, not authenticity or integrity



# Symmetric Cryptography Using Caesar cipher

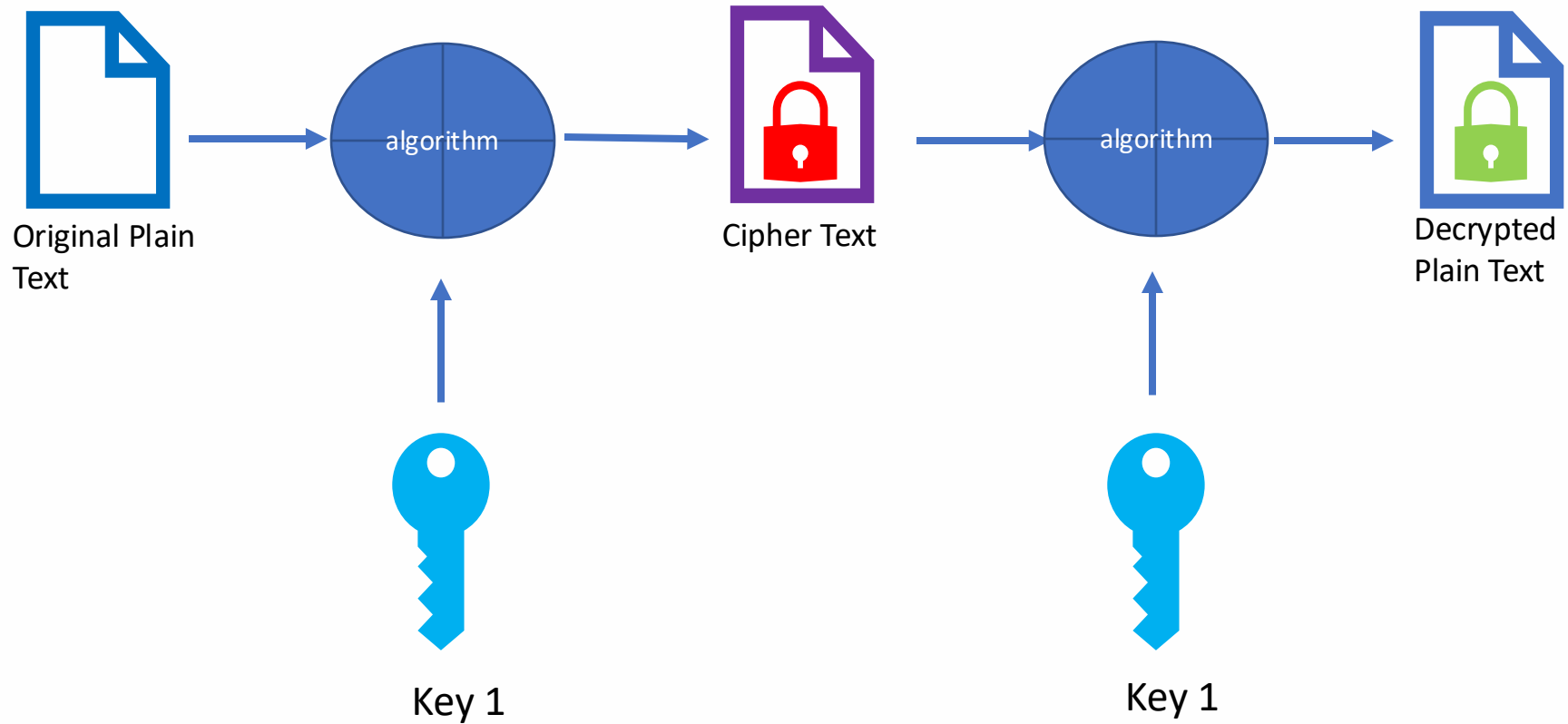
- ✓ A letter in the plain text is replaced with another letter in a fixed order in the alphabet
- ✓ The fixed number of positions is a key both for encryption and decryption
- ✓ This is the earliest form of Substitution Cipher



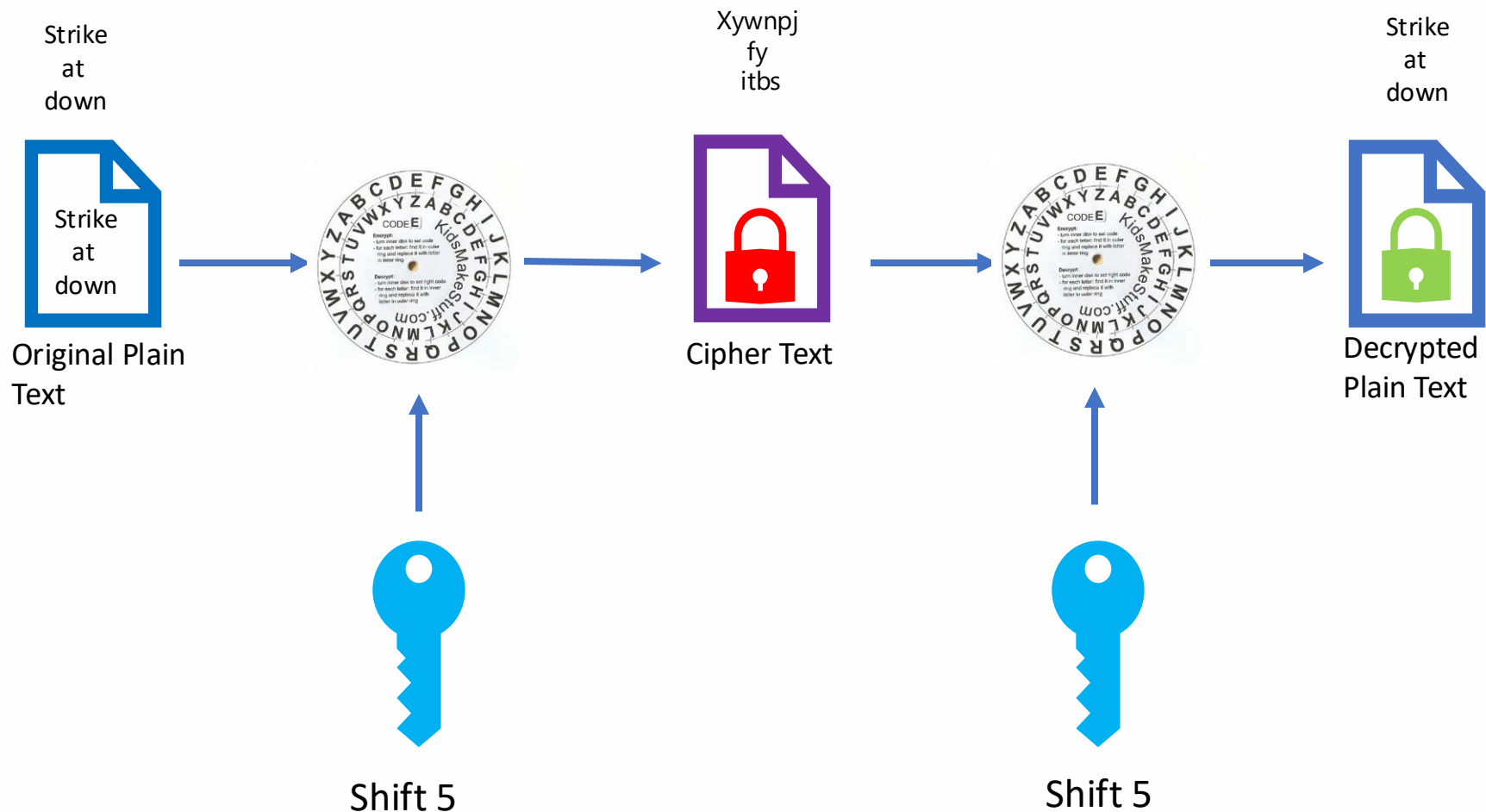
Hubert Berberich ([HubiB](#))



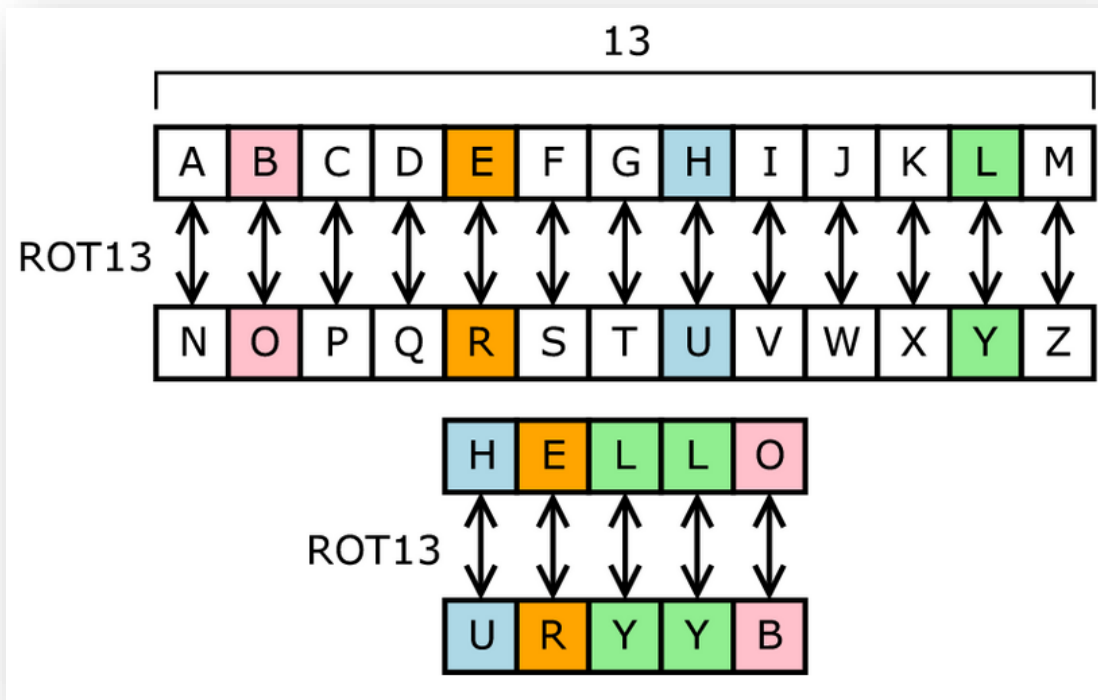
# Symmetric Key



# Symmetric Cryptography Using Caesar Cipher



# Substitution cipher



- ✓ The method of encoding by which units of plaintext are replaced with another plaintext according to a fixed system
- ✓ The "units" may be single letters (the most common), pairs of letters, triplets of letters, mixtures of the above
- ✓ Used in Symmetric algorithms
- ✓ Provides confusion to cryptanalyst





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# Transposition cipher

- ✓ Transposition cipher, also called columns permutation, is a technique to change the order of the letters in a text by placing it in a grid
- ✓ Scytale is a transposition cipher
- ✓ A band is wrapped around a rod, a message is written, and when the band is unrolled a ciphertext appears

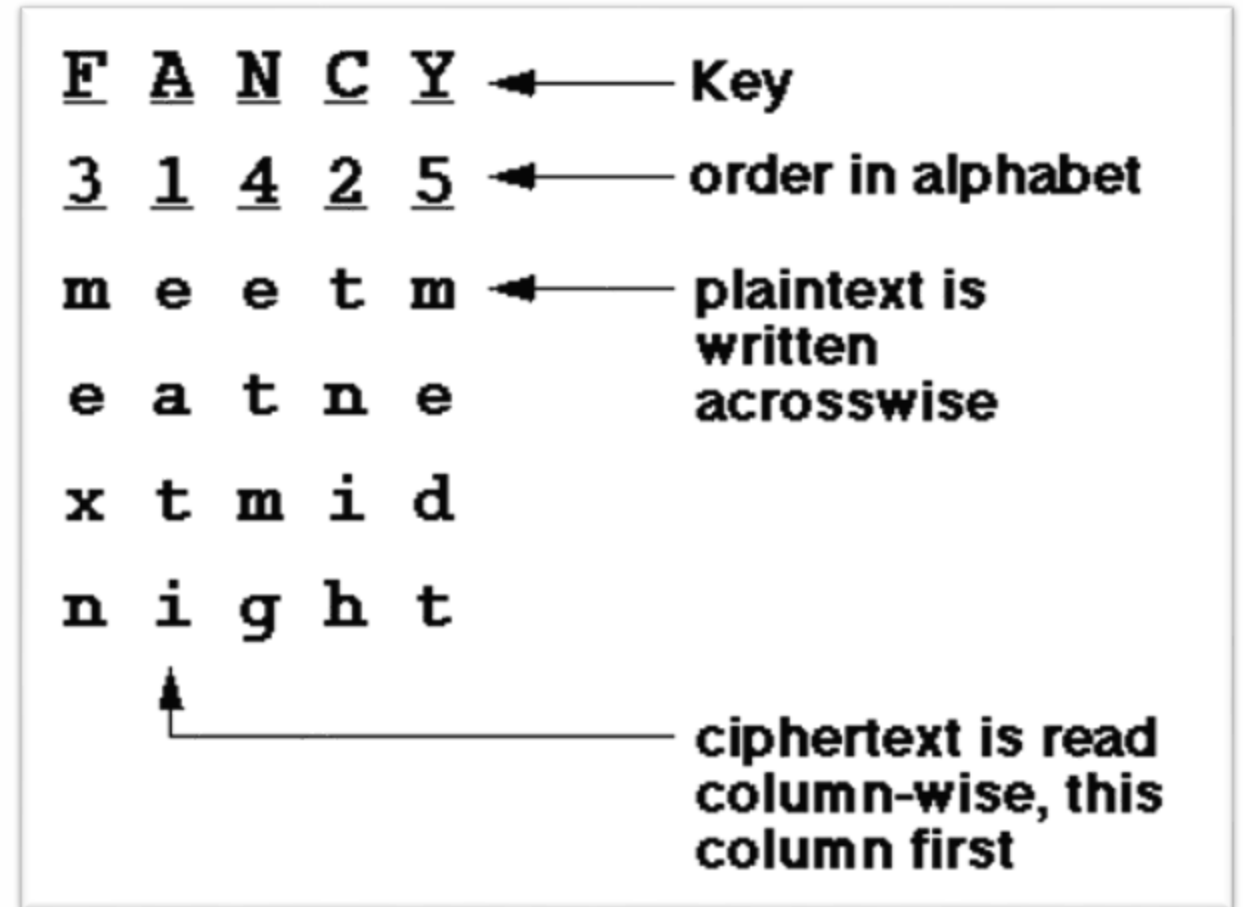


Wikipedia image by: [User:Chrkl](#)



# Transposition Ciphers

Transposition Cipher also called columns permutation, is a technique to change the order of the letters in a text by placing it in a grid



# Vigenère cipher

Plaintext: DIVERTTROOPSTOEASTRIDGE

Keyword: WHITEWHITEWHITEWHITEWHI



|    | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1  | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A |
| 2  | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B |
| 3  | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C |
| 4  | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D |
| 5  | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E |
| 6  | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F |
| 7  | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G |
| 8  | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H |
| 9  | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I |
| 10 | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J |
| 11 | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K |
| 12 | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L |
| 13 | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M |
| 14 | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
| 15 | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O |
| 16 | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
| 17 | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
| 18 | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R |
| 19 | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S |
| 20 | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
| 21 | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |
| 22 | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V |
| 23 | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W |
| 24 | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |
| 25 | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y |
| 26 | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |



vijfhart  
IT-OPLEIDINGEN

Blaise de Vigenère

**Dat klopt voor jou!**

[https://www.simonsingh.net/The\\_Black\\_Chamber/v\\_square.html](https://www.simonsingh.net/The_Black_Chamber/v_square.html)

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# Exercise 1

**Cipher brain workout (25 mins)**



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# Cryptosystem Strength



The *strength* of an encryption method depends on the:

- Algorithm
- The secrecy of the key
- The length of the key
- The initialization vectors
- How everything works together



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# Work Factor



An estimate of the effort and resources it would take an attacker to penetrate a cryptosystem



# Modern Cryptography



- ✓ Works with Binary Data
- ✓ All data must be in Binary
  - ✓ Plaintext
  - ✓ Ciphertext
  - ✓ Key



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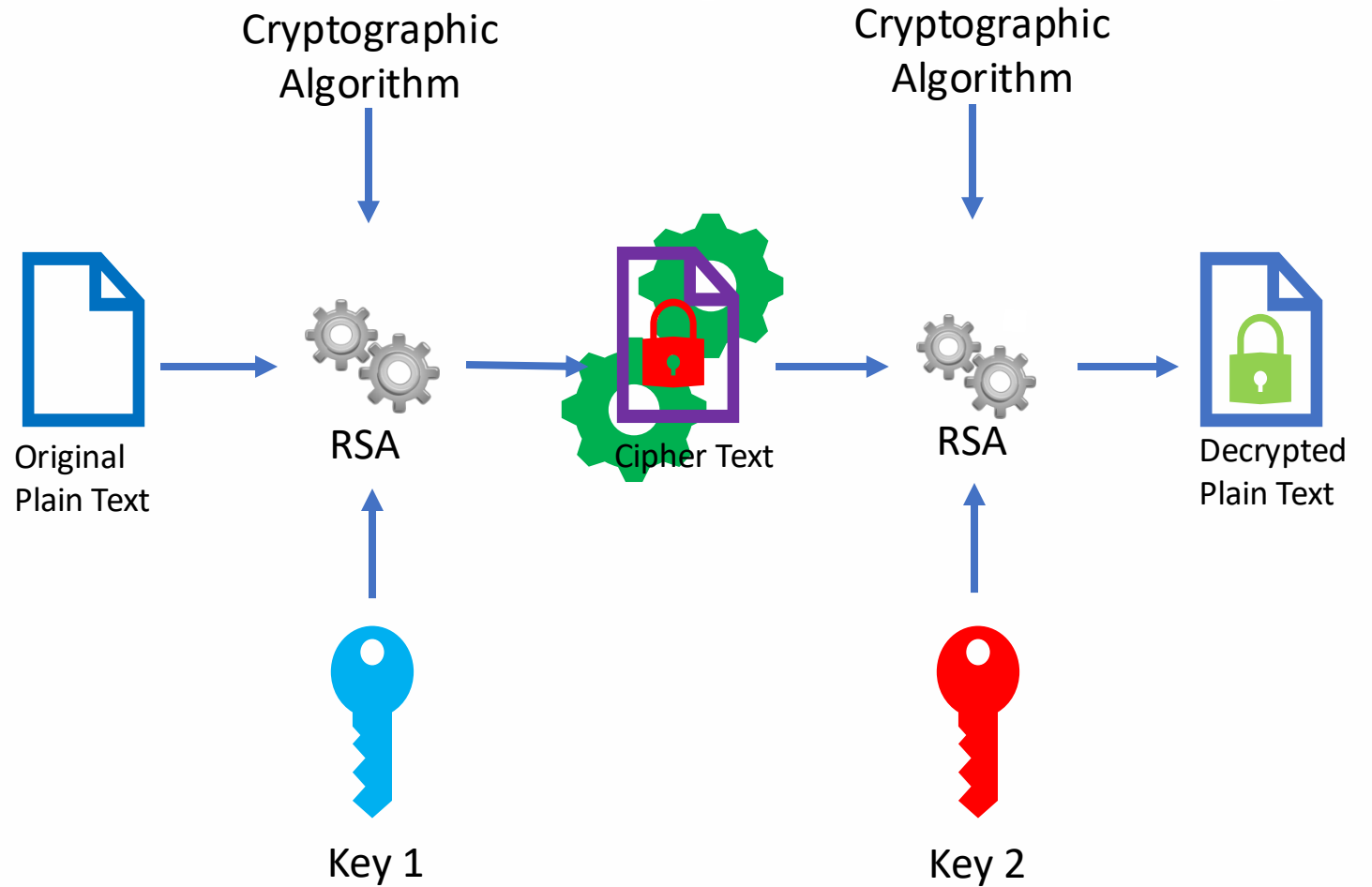
# Asymmetric Cryptography

- ✓ Asymmetric Cryptography makes use of TWO keys
- ✓ The mathematically related pair of keys are called **Private and Public key**
- ✓ Each key can be used for encryption and decryption
- ✓ However, the same key can not encrypt/decrypt the same document
- ✓ Public key is usually shared while private key is secured by the owner





When you encrypt information with one key and decrypt it with another key we call that Asymmetric Keys (2 keys) or Asymmetric cryptography



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# Asymmetric Cryptography

## Advantage

- ✓ Scalable
- ✓ Provides confidentiality, authentication and non- repudiation
- ✓ Better key distribution mechanism
- ✓ Examples: RSA, ECC, DH, El Gamal, DSA

## Disadvantage

- ✓ Slower than symmetric key
- ✓ Mathematically complex and intensive task

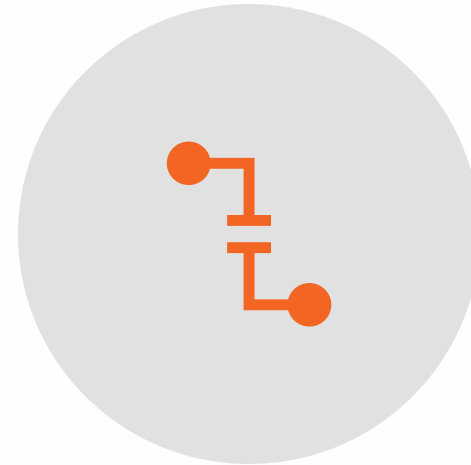


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# Symmetric Algorithm Types



BLOCK CIPHERS – WORKS ON BLOCKS  
OF BITS

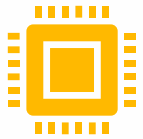


STREAM CIPHERS – WORKS ON ONE  
BIT AT A TIME



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# Stream Cipher



Each plaintext digit is encrypted one at a time with the corresponding digit of the keystream



Uses a keystream generator ~ produces a stream of bits that is X-ored with the plain text stream



It is better suited to be implemented at the hardware level



Characteristics of a strong stream cipher



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# Stream Cipher Pros & Cons

## Advantages

- ✓ Stream ciphers can encrypt and decrypt more quickly
- ✓ Can scale better within increased bandwidth requirements
- ✓ Used when real-time applications, as in VoIP or multimedia are encrypted

## Disadvantages

- ✓ Considered less secure than block ciphers
- ✓ Requires a lot of randomness and encrypt individual bits at a time
- ✓ Requires more processing power than block ciphers
- ✓ Generating a truly random and unbiased keystream is difficult



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# Block Cipher

- ✓ Plaintext is divided into blocks of data and is encrypted one block at a time
- ✓ Most block ciphers work with 32-, 64- or 128-bit block sizes
- ✓ Uses both confusion and diffusion to strengthen the encryption process
- ✓ Confusion is achieved via substitution and diffusion is achieved via transposition
- ✓ Confusion pertains to making the relationship between the key and resulting ciphertext as complex as possible so the key cannot be uncovered from the ciphertext.
- ✓ Diffusion means that a single plaintext bit has influence over several of the ciphertext bits



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# Hybrid Encryption Methods

- ✓ Use of Symmetric and Asymmetric Algorithms as complementary methods
- ✓ Symmetric key is generally used to encrypt the message
- ✓ Asymmetric key is used to encrypt the Symmetric key
- ✓ This process is more commonly called as digital envelope



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# Session Keys



Single use symmetric key that is used to encrypt/decrypt communication between two users for a single session



It's much more secure than static symmetric keys



Peers decide on the session key and continue to use it till the session is over



Eavesdropping is difficult, breaking the keys is futile



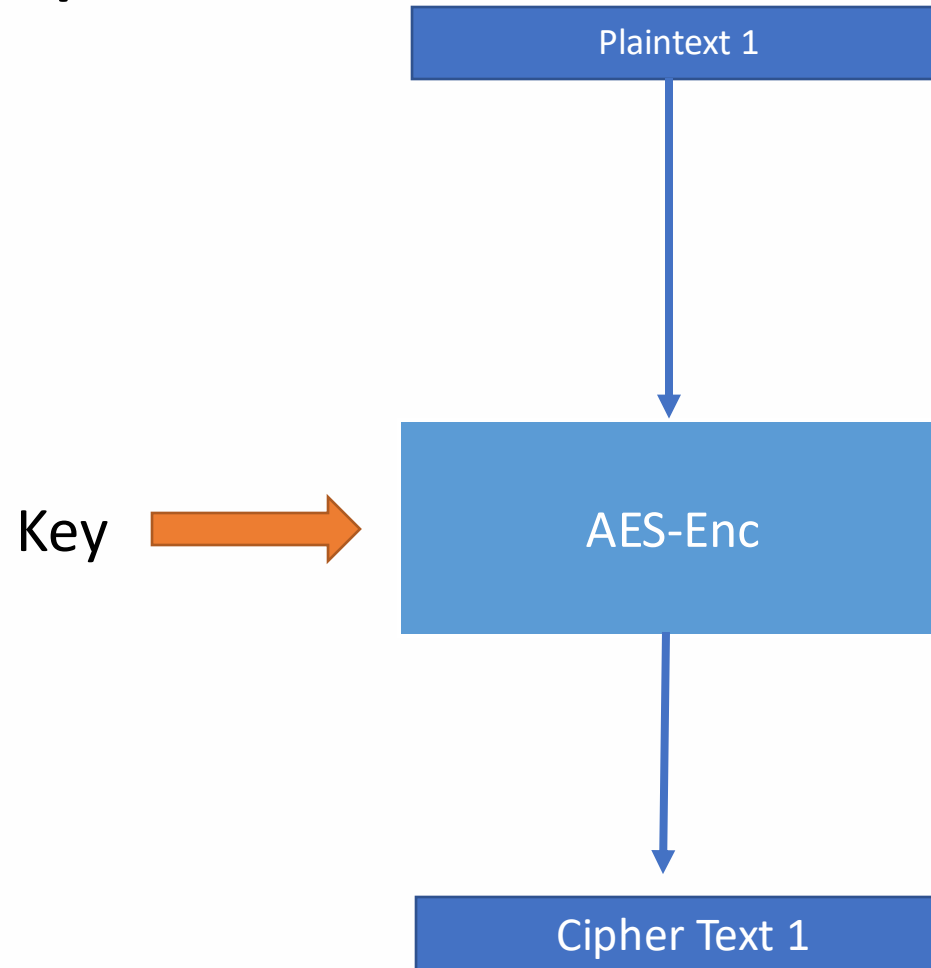


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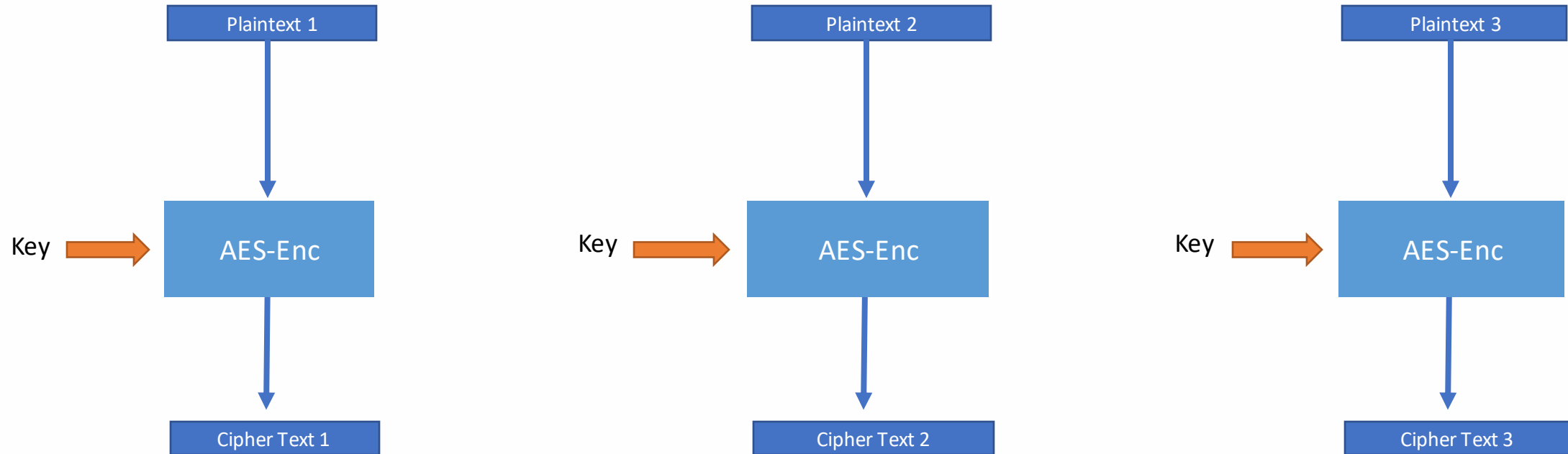
# Cipher Block Mode



# The Block Cipher



# Electronic Code Book (ECB) Mode

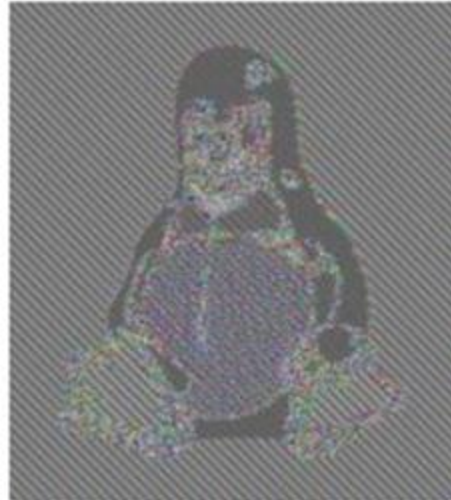


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# The Linux Penguin Problem



Original



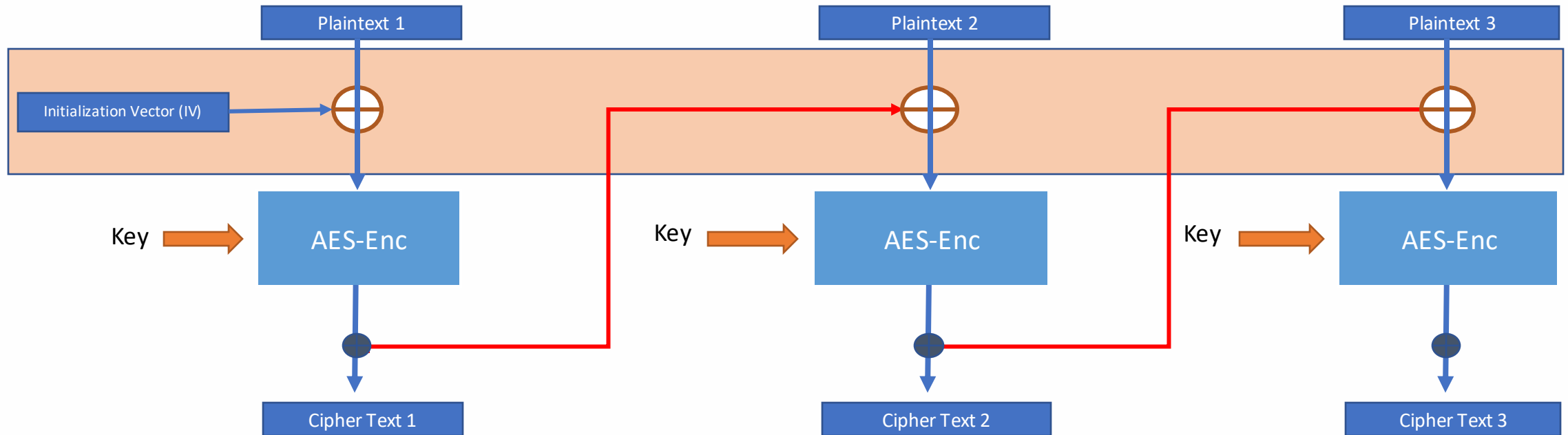
ECB mode



"other" mode



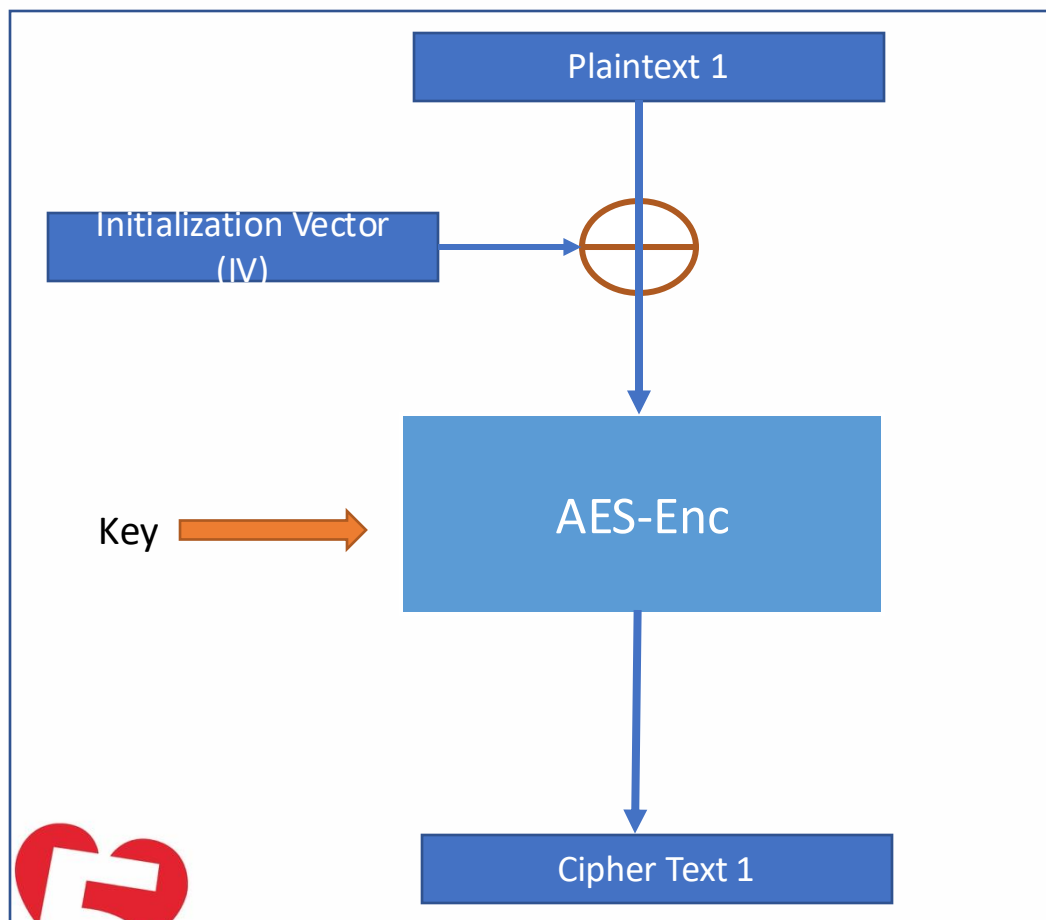
# Cipher Block Chaining (CBC) Mode



- ✓ Solve security deficiencies in ECB
- ✓ Each previous cipher blocks is chained to be input with current plaintext block
- ✓ The encryption of a block depends on the current and all blocks before it.
- ✓ Use for bulk data encryption



# Cipher Block Chaining (CBC) Mode



The Initialization Vector is a random values that is used with algorithms to ensure patterns are not created during the encryption process



They are used with keys and do not need to be encrypted when being sent to the destination



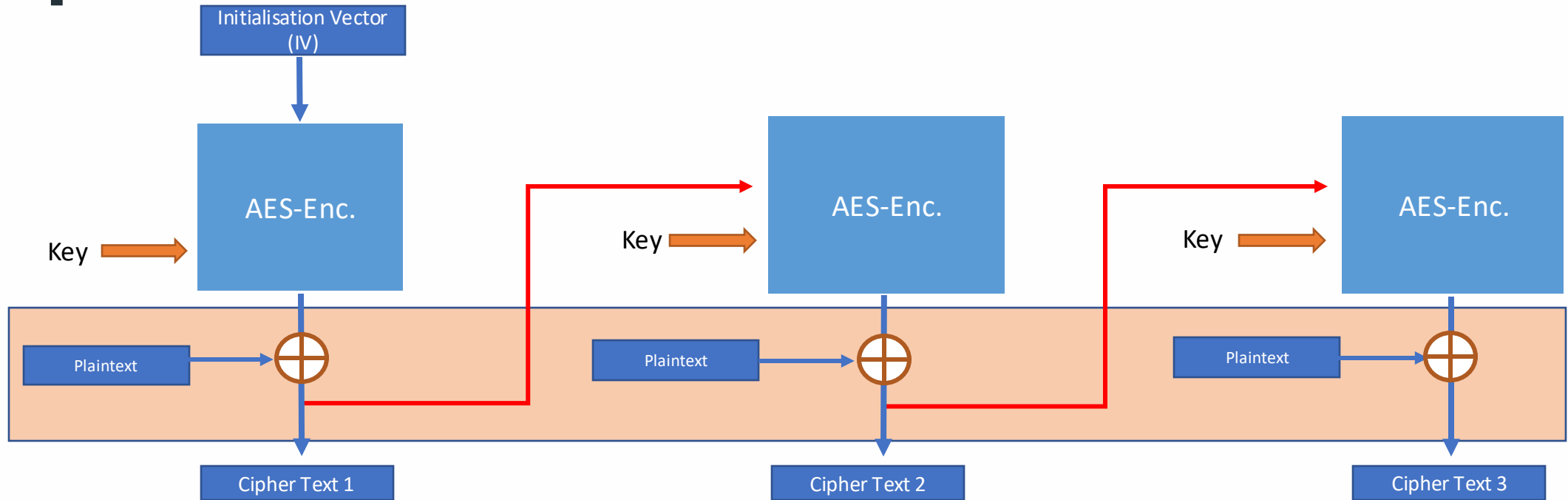
If IVs are not used, then two identical plaintext values that are encrypted with the same key will create the same ciphertext.



The IV and key are both used by the algorithm to provide more randomness to the encryption process



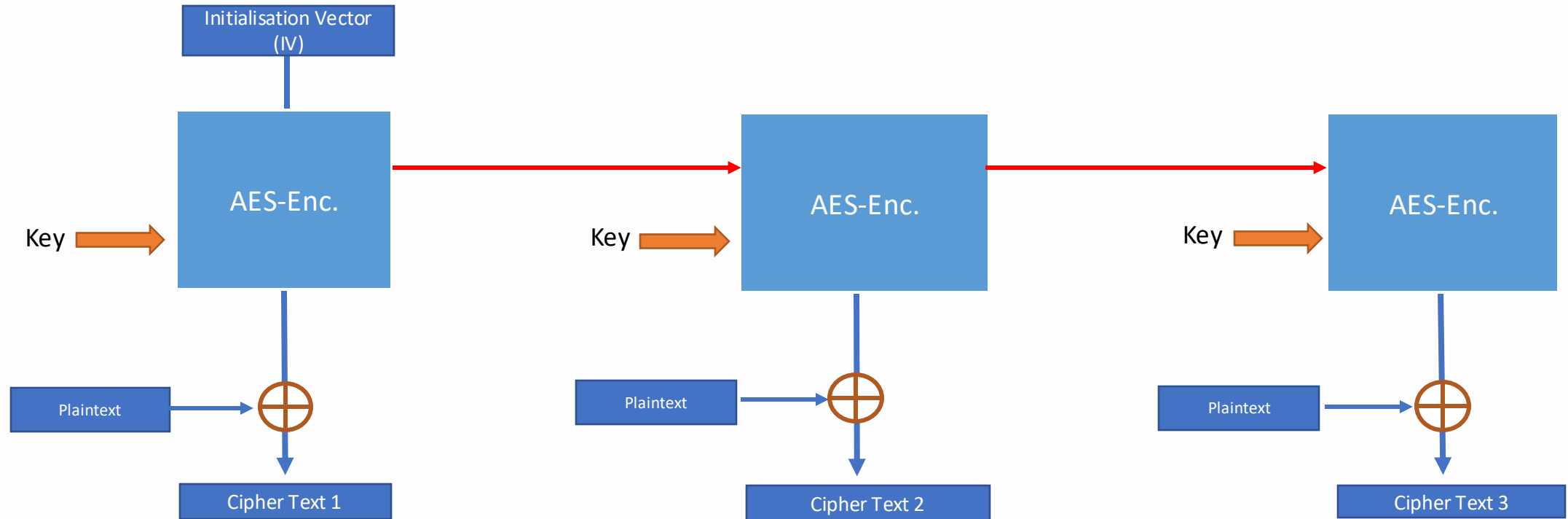
# Cipher Feedback Mode (CFB) Mode



- ✓ The block cipher is used as a stream cipher, and can encrypt any number of bits e.g. single bits or single characters (bytes)



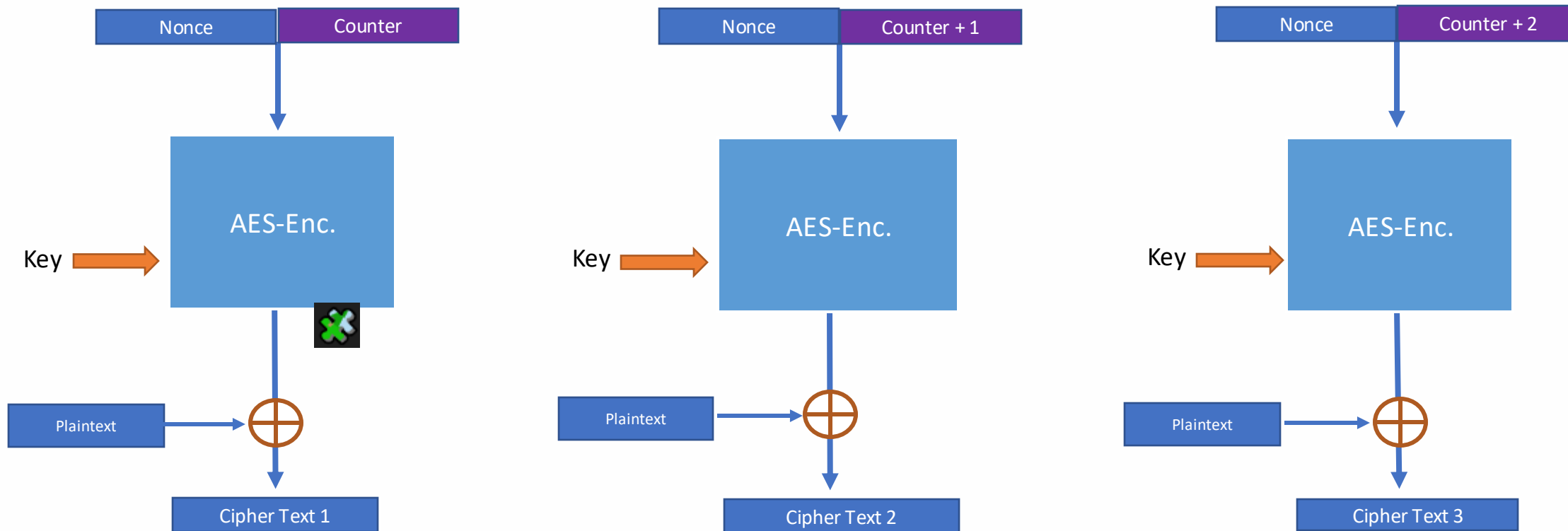
# Output Feedback Mode (CFB) Mode



- ✓ The block cipher is used as a stream cipher for digital video, audio signals.
- ✓ Used when data arrives in bits/bytes and when error propagation is not accepted
- ✓ Resistant to transmission errors; a bit error in a ciphertext segment affects only the decryption of that segment.



# Counter (CTR) Mode



- ✓ Uses an IV counter that increments for each plaintext block that needs to be encrypted
- ✓ Produces a sequence which is guaranteed not to repeat for a long time
- ✓ Counter must be unknown and unpredictable
- ✓ Uses: high-speed network encryptions, Encrypting ATM cells, IPsec, Wireless 802.11i

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# Exercise 2

**XOR**



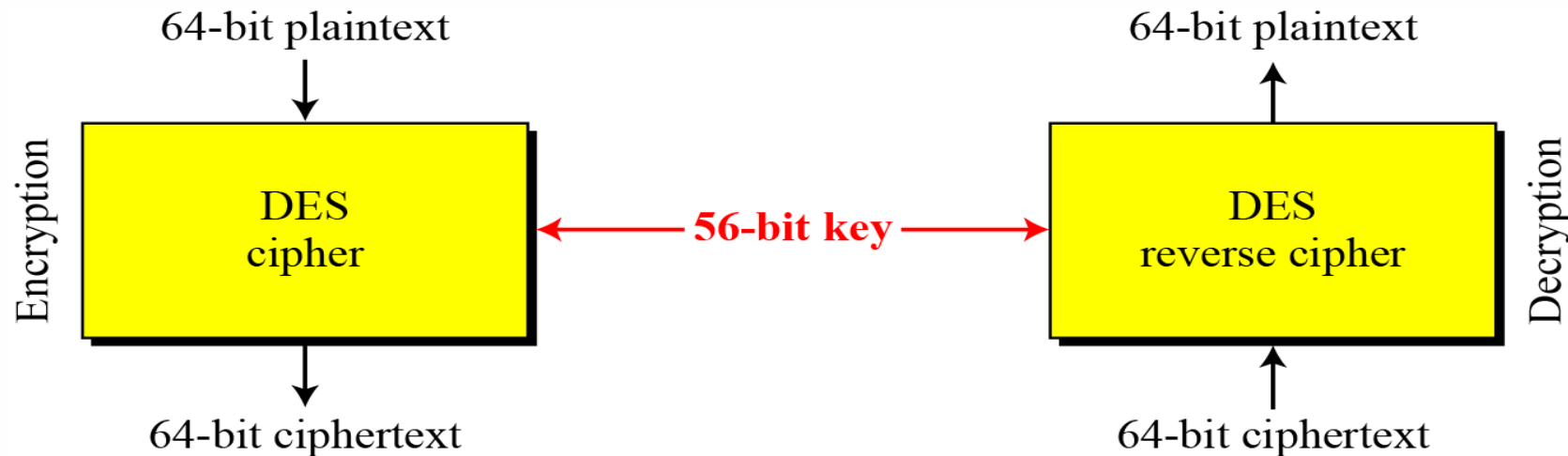
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# Cryptographic Algorithm



# The Data Encryption Standard (DES)

- ✓ The Data Encryption Standard (DES) is a symmetric-key block cipher
- ✓ It undergoes 16 rounds of substitution and transposition functions



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# The Advanced Encryption Standard (AES)

- ✓ The Advanced Encryption Standard (AES) is a symmetric-key block cipher published by the National Institute of Standards and Technology (NIST) in December 2001
- ✓ Block sizes: 128 bits
- ✓ Key sizes: 128, 192 or 256 bits
- ✓ Rounds: 10, 12 or 14 (depending on key size) It uses 10, 12, or 14 rounds. The number of rounds depends on the key size



# Rijndael

- ✓ Rijndael is also the Advanced Encryption Standard (AES), BUT with a Variable Block size and key length
  - ✓ Block size: 128, 192, 256
  - ✓ Key length: 128, 192, 256
- ✓ 9 rounds are used if the key or block size is 128 bits.
- ✓ 11 rounds of transformation are deployed if the key or block size is 192 bits.
- ✓ 13 rounds are involved if the key or block size is 256 bits.



Vincent Rijmen



Joan Daemen



# The International Data Encryption Algorithm (IDEA)

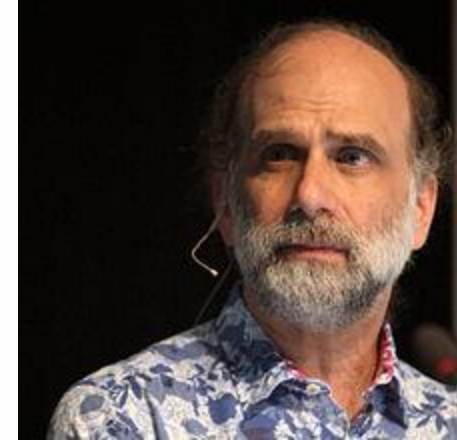
- ✓ IDEA is a block cipher
- ✓ IDEA operates with 64-bit plaintext and cipher text blocks and is controlled by a 128-bit key
- ✓ The 64-bit plaintext block is partitioned into 16-bit sub-blocks, each has 8 rounds of mathematical functions performed on it.
- ✓ IDEA is faster than DES when implemented in Software used in PGP
- ✓ It is a patented algorithm



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# Blowfish

- ✓ Block cipher: 64-bit block
- ✓ Variable key length: 32 bits to 448 bits
- ✓ Data blocks go through 16 rounds of cryptographic functions
- ✓ Much faster than DES and IDEA



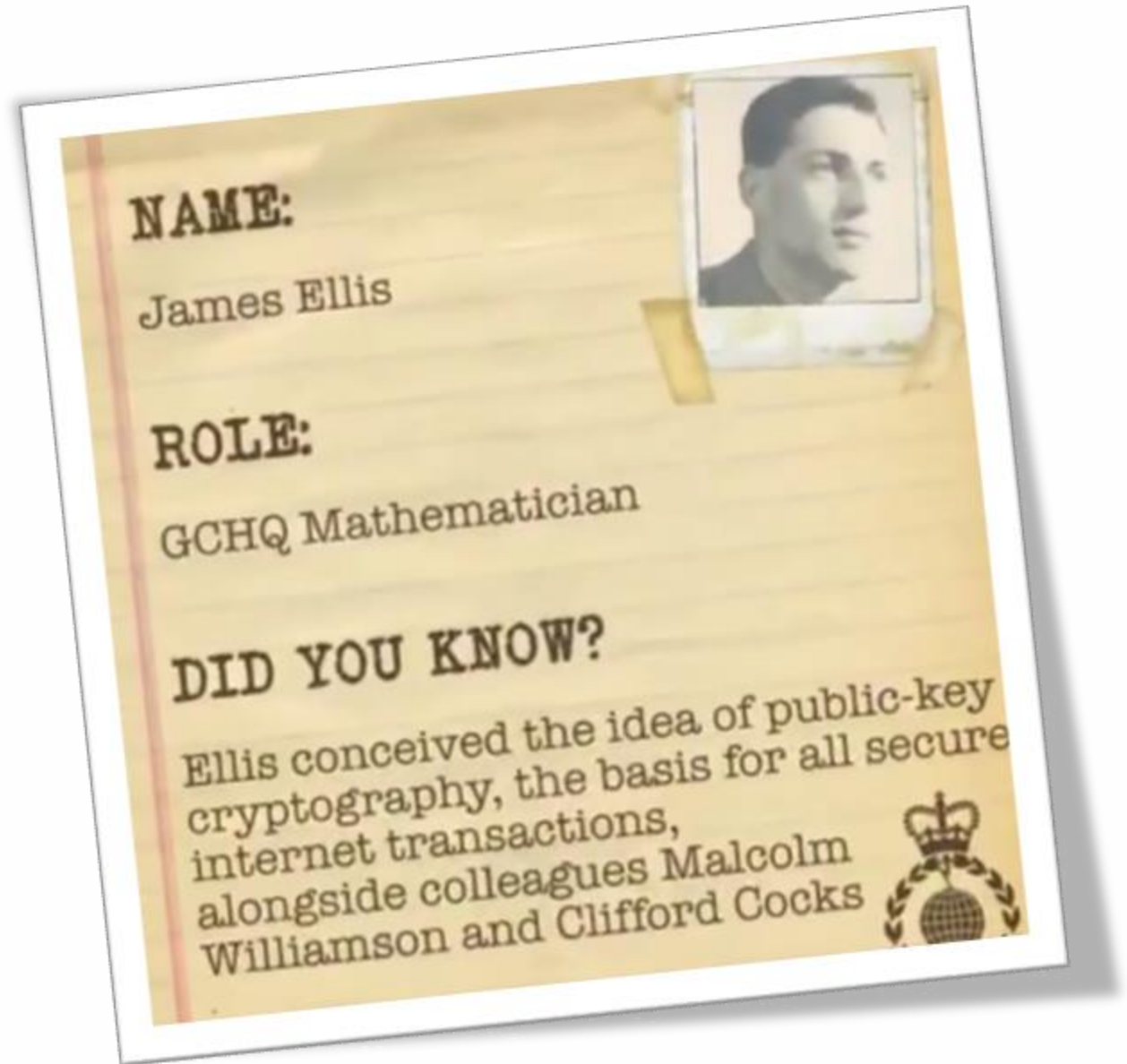
**Bruce Schneier**





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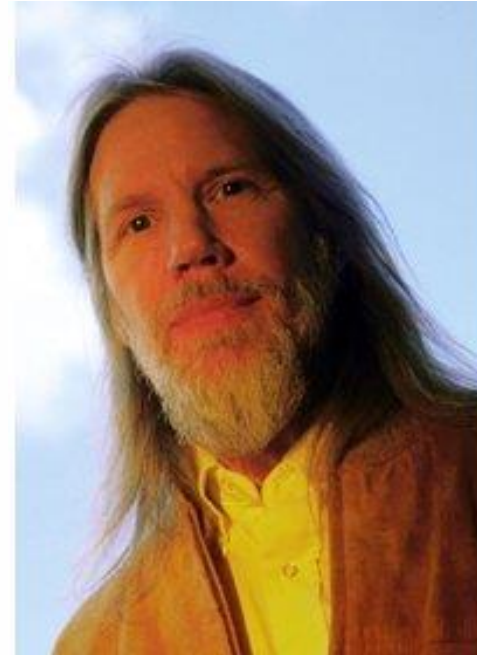
# Public Key Cryptography



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# Diffie-Hellman (DH)

- ✓ Diffie-Hellman is the key distribution asymmetric algorithm widely used today to distribute symmetric keys
- ✓ Allows two users to exchange a secret key without requiring prior secrets
- ✓ Does not provide for encryption or digital signature functions

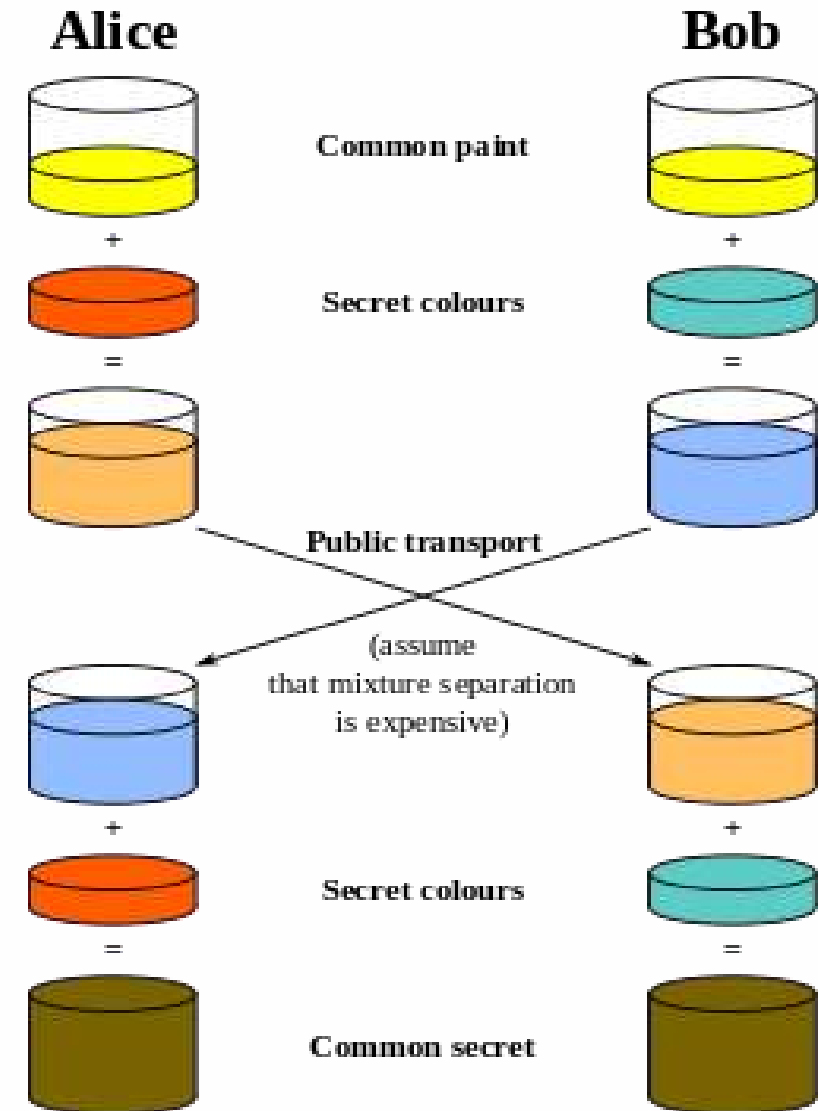


Authors of the first published paper on Public-key cryptography.



# Diffie-Hellman (DH)

- ✓ Currently used in many protocols, namely:
  - ✓ Secure Sockets Layer (SSL)/Transport Layer Security (TLS)
  - ✓ Secure Shell (SSH)
  - ✓ Internet Protocol Security (IPSec)
  - ✓ Public Key Infrastructure (PKI)



# Rivest, Shamir and Adleman (RSA)



[Ron Rivest, Adi Shamir and Leonard Adleman.](#)

*(Wikimedia Commons)*

- ✓ Most popular asymmetric algorithm used for key exchange, encryption, digital signature and authentication
- ✓ Based on the difficulty of factoring large semi prime number which is a product of two large prime numbers
- ✓ The result is a number that can only be broken down into the initial prime numbers
- ✓ The public and private keys are a function of large prime numbers

<https://www.youtube.com/watch?v=ZKKDTfHcsG0>



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**Dat klopt voor jou!**



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# RSA

- ✓ A prime number is a number that is only divisible by 1 or by itself
- ✓ The first few primes are 2, 3, 5, 7 and 11. The rest, like 4 for instance, are not prime: 4 can be broken down to 2 times 2, as well as 4 times 1.
- ✓ Using one-way it provides encryption and signature verification, reverse way it provides decryption and signature generation

406,541,318,870,452,164,142,161,028,497,034,055,908,330,795,850,575,  
741,649,307,604,240,081,597,502,233,885,902,517,289,285,023,585,791,  
223,239,592,783,051,318,295,484,219,924,025,332,854,246,132,728,715,  
552,460,619,944,194,679,330,033,676,282,829,328,152,739,714,673,366,  
179,906,823,029,877,973,507,929,239,423,443,377,132,384,753,724,178,  
447,226,041,912,767,138,249,925,309,993,357,379,153,572,212,166,426,  
773,538,079,561,418,021,293,020,569,682,116,684,933,148,131,688,925,  
508,523,929,010,548,063,794,858,755,212,639,973,154,477,995,891,903,  
914,896,767,233,157,948,181,644,383,905,584,704,750,296,848,115,761,  
704,955,875,653,862,778,035,566,296,235,473,942,689,241,421,722,455,  
091,237,726,940,399,605,388,375,173,453,263,419,405,930,790,375,958,  
081,259,142,857,304,624,751,178,998,559,654,155,353,368,257,669,344,  
122,166,097,011,178,766,722,819,809,221,811,483,063,408,587,293,703,  
584,044,846,944,373,009,650,823,628,883,576,287,986,355,136,594,363,



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# RSA Basic Mathematics

- ✓ The RSA algorithm allows a message to be encrypted without the sender knowing the key
- ✓ The public key is generated by multiplying two large prime numbers  $p$  and  $q$  together and the private key is generated through a different process involving  $p$  and  $q$ .
- ✓ Public Key is Generated by : Multiply  $p * q$  to get  $N$
- ✓ Example
  - $P=17$
  - $Q=11$
  - $N= p*q$
  - $N= 17 * 11 = 187$
  - **187 is a part of your public keys**

<https://brilliant.org/wiki/rsa-encryption/>



$$n = p \times q$$

$$N = 76697 \times 454577$$

Prime

Prime

N is easy to calculate



$$n = p \times q$$

$$P = 76697$$

$$Q = 454577$$

$$N = 34864692169$$

Semi Prime





$$n = p \times q$$

$$34864692169 = P \times Q$$

Given N, calculate P & Q. This is a bit more difficult to calculate even for a computer



RSA-170

N

2606262368413984492152987926667443219708592538048640641616  
4785191859999628542069361450283931914514618683512198164805  
919882053057222974116478065095809832377336510711545759

Large Semiprime number



P

35864207304285014867998  
04587268520423291459681  
05997816114023186063394  
8450858 040593963

Prime

X

Q

72670290641070190788637  
97763923946264136137803  
85699667031370893600228  
1582249 587494493

Prime

=

RSA-170

N

2606262368413984492152987926667443219708592538048640641616  
4785191859999628542069361450283931914514618683512198164805  
919882053057222974116478065095809832377336510711545759

Semiprime



P

35864207304285014867998  
04587268520423291459681  
05997816114023186063394  
8450858 040593963

Prime

X

Q

72670290641070190788637  
97763923946264136137803  
85699667031370893600228  
1582249 587494493

Prime

=

RSA-170

N

26062623684139844921529879266674432197085925380486406  
41616478519185999962854206936145028393191451461868351  
21981648059198820530572229741164780650958098323773365  
10711545759

Semiprime



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## What about if this was the product of Two Prime Number....

2855425422282796139015635661021640083261642386447028891992474566022844003906  
00653875954571505539843239754513915896150297878399377056071435169747221107988  
7911982009884775313392142827720160590099045866862549890848157354224804090223  
44297588352526004383890632616124076317387416881148592486188361873904175783145  
6960169195743907655982801885990355784485910776836771755204340742877265780062  
66759615970759521327828555662781678385691581844436444812511562428136742490459  
363212810180276096088111401003377570363545725120924073646921576797146199387619  
29656030268026179011813292501232304644443862230887792460937377301248168167242  
44936744744885377701557830068808526481615130671448147902883666640622572746652  
757871273746492310963750011709018907862633246195787957314256938050730561196775  
8033808433338198750090296883155787878778558758758775878785587878799869587676585  
875875878758758787878787788787878793591309526982131114132239335649017848872898228  
81562826008138312961436638459454311440437538215428712777456064478585641592133  
2844358020642271469491309176271644704168967807009677359042980890961675045292  
725800084350034483162829708990272864998199438764723457427626372969484830475  
09171741861811306885187927486226122933413689280566343844666463265724761672756  
60839105650528975713899320211121495795311427946254553305387067821067601768750  
97786610046001460213840844802122505368905479374200309572209673295475072171811  
5531871310231057902608580607



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## Imagine if this is one of our Prime Factors

The Largest prime number  
is 23,249,425 million  
digits.

A number big enough to fill an entire shelf  
of books totaling 9,000 pages.



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# Integrity Authentication And Access Control

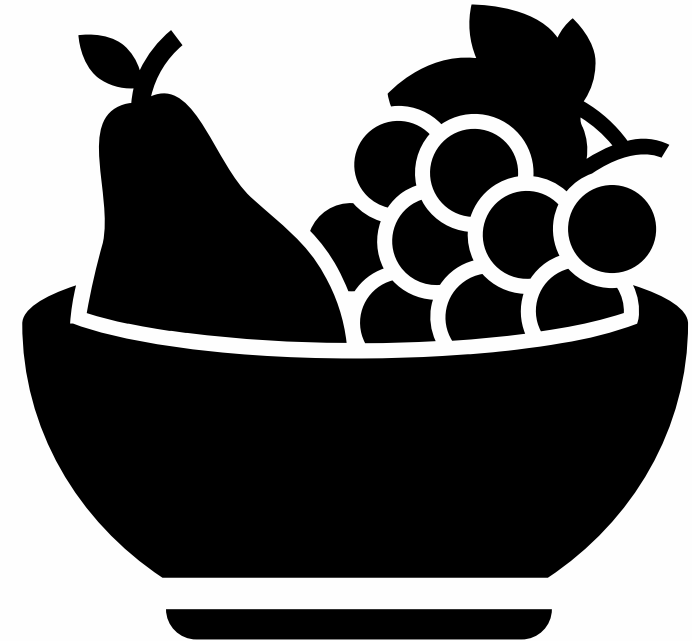


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# Hashing

- ✓ Encryption (RSA, AES) is reversible with the key
- ✓ Hashing is one way only, we can't go back to the original data

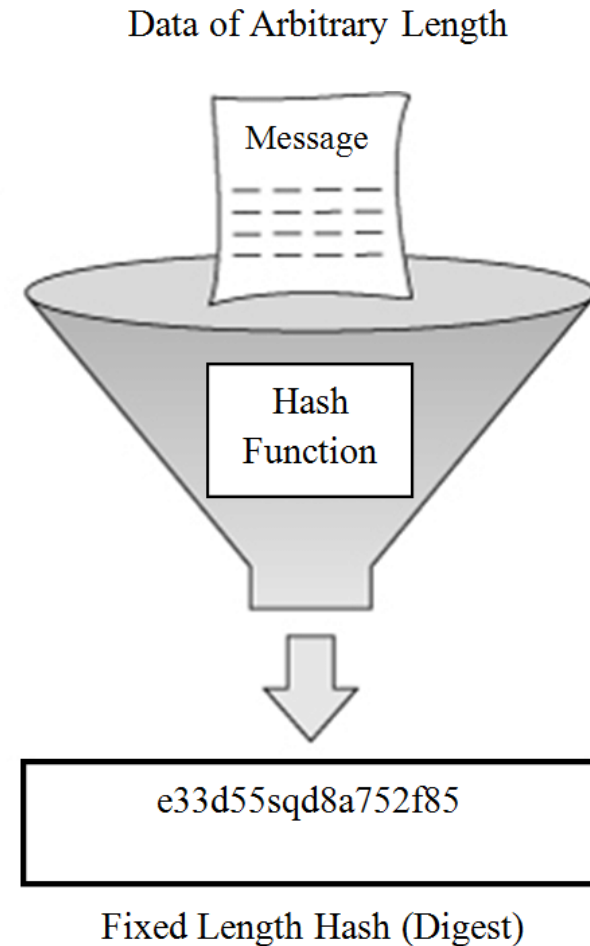
Think of smoothies! Can you unblend a smoothie?





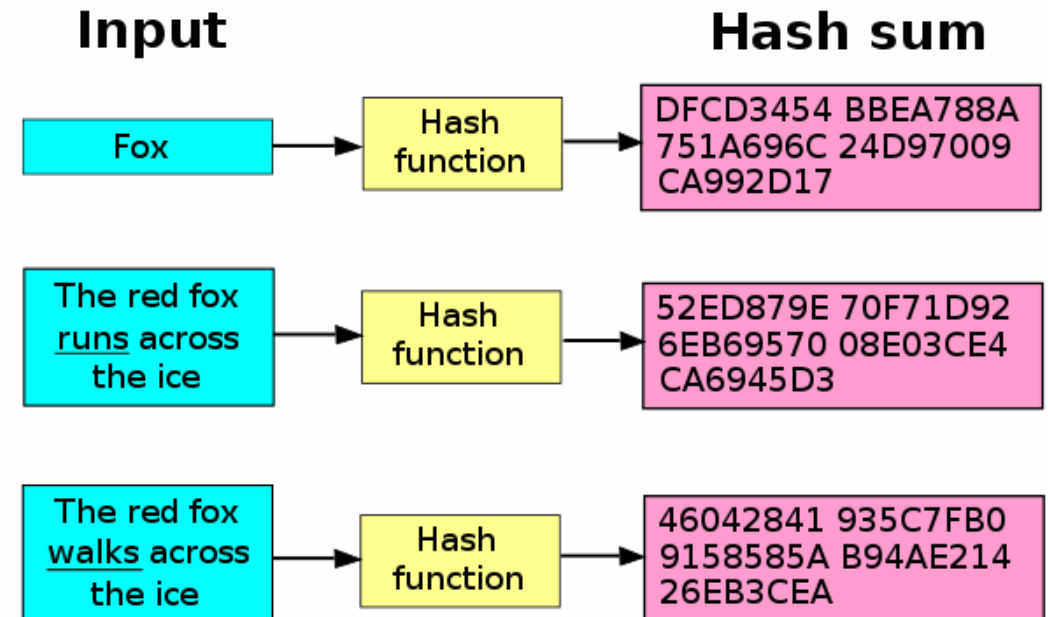
# Data Hashing

- ✓ Data of arbitrary size is feed through a hashing algorithm to produce a fixed-size value
- ✓ The value returned by a hash algorithm is called hash
- ✓ What can you do with Hashing?
  - ✓ Data integrity
  - ✓ Secure passwords storage in a database
  - ✓ Authentication



# One-way Hash function

- ✓ A Hash is used to guarantee the integrity of data
- ✓ A Hash take a single input – a message and produces a message digest
- ✓ Hash can be applied to any size data block
- ✓ Hash produces fixed-length output
- ✓ One-way hash function is never used in reverse



# Characteristics of Strong Hashing function

- ✓ Deterministic - must always provide output of a fixed length no matter the size of the message
- ✓ Hash should be computed over the entire message
- ✓ Hash should be a one-way function - Pre-Image Resistance
- ✓ Collision resistance - Given a message and Hash value, computing another message with the same Hash value should be impossible
- ✓ Speed – Hashing should be fast
- ✓ Any slight change made to the input should result in a massive change in the fixed length output - Avalanche Effect



# Hash Results

Hashing is a method of cryptography that converts any form of data into a unique string of text. Any piece of data can be hashed, no matter its size or type. In traditional hashing, regardless of the data's size, type, or length, the hash that any data produces is always the same length.



|           |                                          |
|-----------|------------------------------------------|
| SHA1 Hash | 54a0d99004385c75206b17b1821924295f1ff60b |
| MD5 Hash  | 3f70da9b27380c480c35cac56a9daae7         |

Hashing is a method of cryptography that converts any form of data into a unique string of text. Any piece of data can be hashed, no matter its size or type. In traditional hashing, regardless of the data's size, type, or length, the hash that any data produces is always the same length.

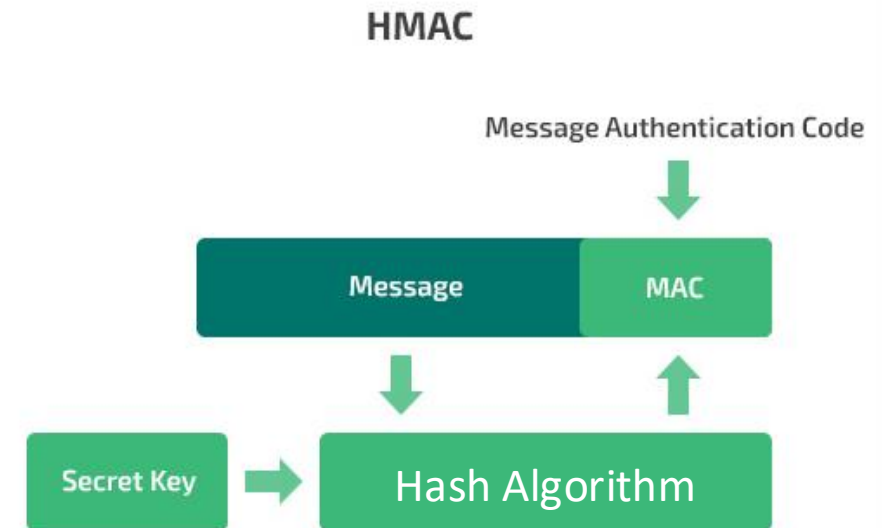


|           |                                         |
|-----------|-----------------------------------------|
| SHA1 Hash | f26eca8e13a5ca4b92e1d29c1c0232a9d42009c |
| MD5 Hash  | 3387bd006987045bae3d15cead19acad        |



# Hash-based Message Authentication Code (HMAC)

- ✓ Uses cryptographic hash function in combination with a secret key to verify both the data integrity and the authenticity of a message via a communication channel
- ✓ The Symmetric key is concatenated with the message before it is sent to Hashing algorithm; the output is the MAC value that is appended to the message and sent
- ✓ The receiver must have the same symmetric key to concatenate and run the Hashing algorithm to obtain the matching MAC value
- ✓ Message is not encrypted with HMAC, hence no confidentiality provided



# Cracking Hash with Rainbow Table

- ✓ Rainbow tables of reversed hashes used to crack password hashes
- ✓ They are massive pre-computed lookup tables containing words, password, and their corresponding hashes
- ✓ This makes it easy to lookup a password hash instantly without the need to run it through a crack

| Hash                                                            | Password                         |
|-----------------------------------------------------------------|----------------------------------|
| 870285B564C496853BDC137C6472995A                                | Amsterdam)))!                    |
| B1543717C8612898D59D8D429AC03A8F                                | Money3DJ                         |
| gdN%Sh35Cj4d#6cH                                                | 417663E9BABDE35116C16A8DE4428DBC |
| ThisIsAveryLongPasswordLikeProfessionalWANTUStouse_Sh35Cj4d#6cH | 7BCBCDB0A5AE6C952685439354C5BF29 |



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# SSL/TLS



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# Security at the Transport Layer

## Secure Socket Layer (SSL)

- Developed by Netscape to provide security in WWW browsers and servers
- SSL is the basis for the Internet standard protocol
  - An SSL session is an association between two peers
  - An SSL connection is the set of mechanisms used to transport data in an SSL session





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# Why Use SSL/TLS

## → Encryption

→ SSL/TLS encrypt the data transmitted across the web so anybody intercepting it cannot read it

## → Authentication

→ The parties involved in any communication is authenticated and verified that they are who they say they are

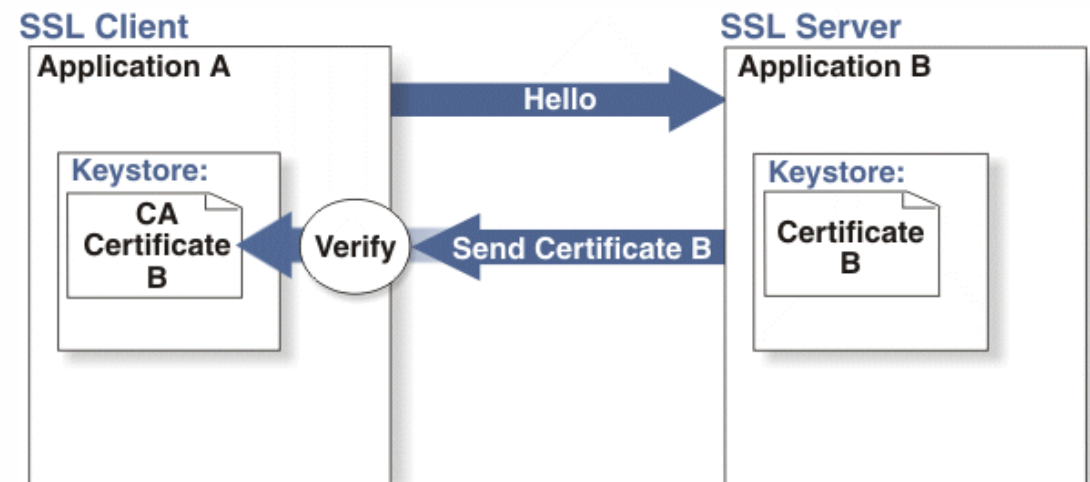
## → Integrity:

→ The data is digitally signed to provide data integrity



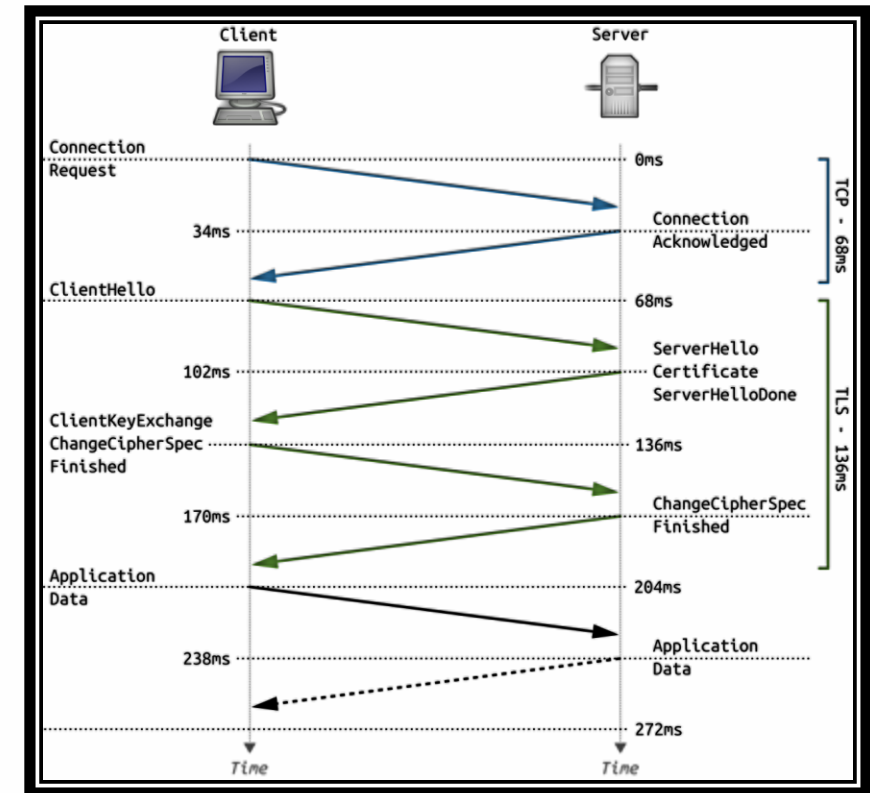
# How SSL/TLS works

SSL initiates an **authentication** process called a handshake between two communicating devices to ensure that both devices are really who they claim to be



# What Happens During SSL/TLS Handshake

- During the Handshake, server and client exchange required information to establish a secure connection.
- Specify which version of TLS (TLS 1.0, 1.2, 1.3) they will use
- Decide on which cipher suites (encryption algorithm) to use
- **Authenticate** the identity of the server using the server's public key and the SSL certificate authority's digital signature
- Generate session keys in order to use symmetric encryption after the handshake is complete



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# SSL/TLS Handshake Protocol

- Two Type of SSL Handshake
  - One-way SSL
  - Two-Way SSL (Mutual SSL)



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# Kerberos

## Authentication Protocol



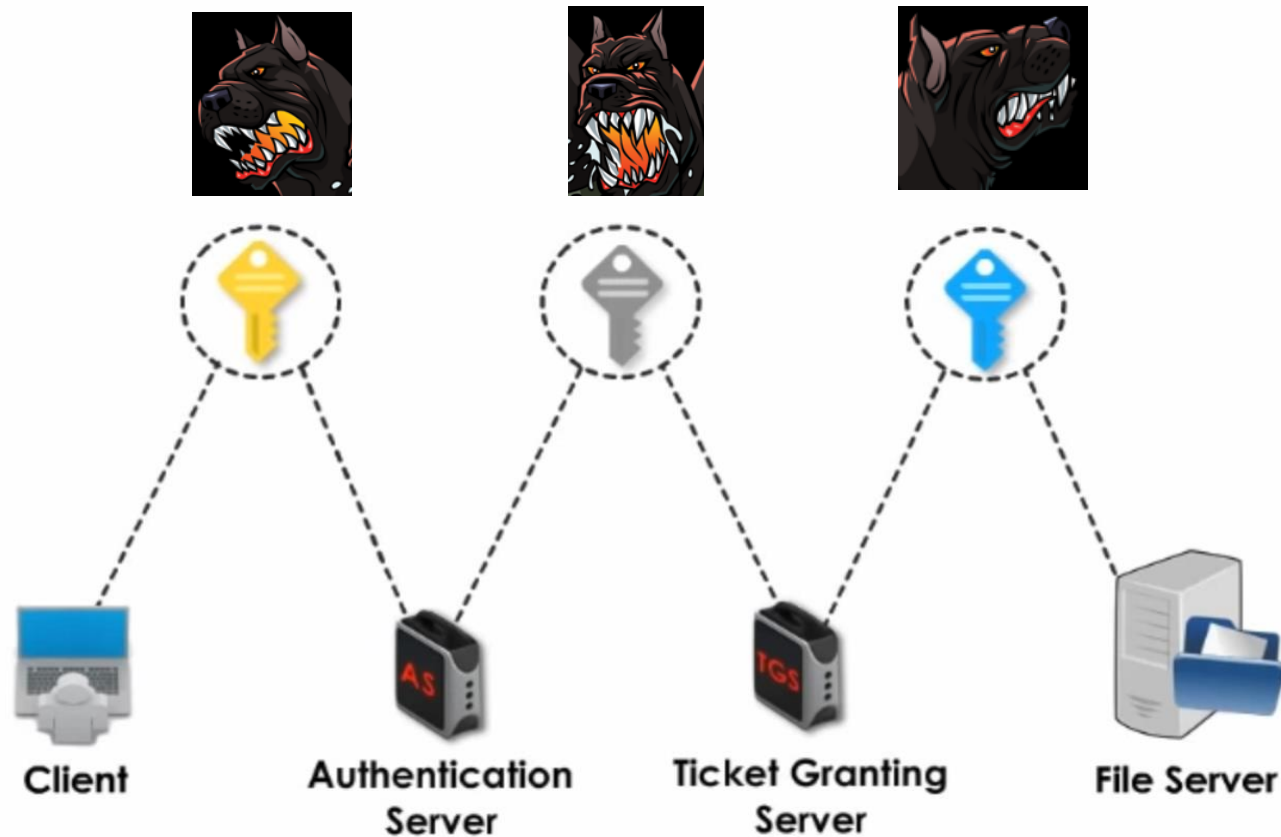
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# Authentication With Kerberos

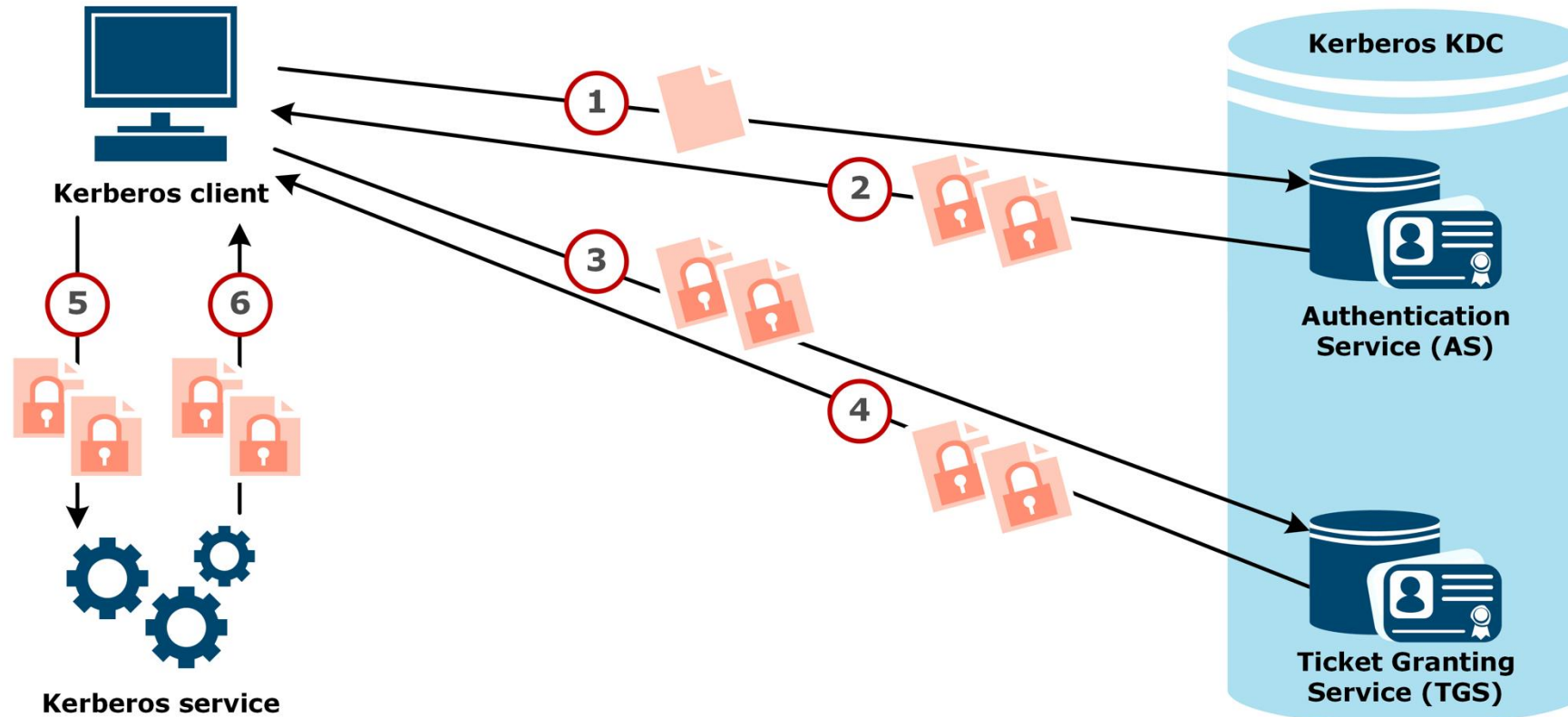
- Kerberos is a de facto authentication standard for heterogeneous networks and used in distributed environments
- It's an authentication protocol that uses Symmetric key algorithm (but it's not an encryption algorithm itself)
- It provides end-to-end security
- Most Kerberos authentications work with shared secret keys, it eliminates the need to share the passwords over the network



# The Three Headed Dog



# How Kerberos Works





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# Kerberos - components

- **Authentication Server**

- Hosts the functions of the KDC: Authentication service (AS) and a Ticket granting Service (TGS)

- **Ticket Granting Ticket (TGT)**

- Provides proof that a subject has authenticated through a KDC and is authorized to request tickets to access other objects
  - TGT is encrypted and includes a symmetric key, an expiration time and the user's IP address
  - Subjects present the TGT when requesting access to objects Ticket
  - Is an encrypted message that provides proof that a subject is authorized to access an object
  - It is sometimes called a Service Ticket (ST)



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# Public Key Infrastructure (PKI)



# Public Key Infrastructure (PKI)



ISO Authentication framework  
with two components

Asymmetric algorithm  
X.509 standard



Setup to handle authentication across networks



It provides Authentication, Confidentiality, Access control,  
Integrity and non-repudiation



---

# Public Key Infrastructure (PKI)

- PKI is made up of
  - Public Key
  - Private Key
  - Root CA
  - Intermediate CA
  - Registration Authority
  - Certificate Authority
  - Certificate Revocation List



# Private, and Public Key Pair



public



private

----- BEGIN PUBLIC KEY BLOCK -----

----- BEGIN PRIVATE KEY BLOCK -----

-----BEGIN PUBLIC KEY BLOCK-----

```
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA...
-----END PUBLIC KEY BLOCK-----
```

-----BEGIN PRIVATE KEY BLOCK-----

```
MIIEvQIBADANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA...
-----END PRIVATE KEY BLOCK-----
```

----- END PUBLIC KEY BLOCK -----

----- END PRIVATE KEY BLOCK -----

Public Key  
Private Key

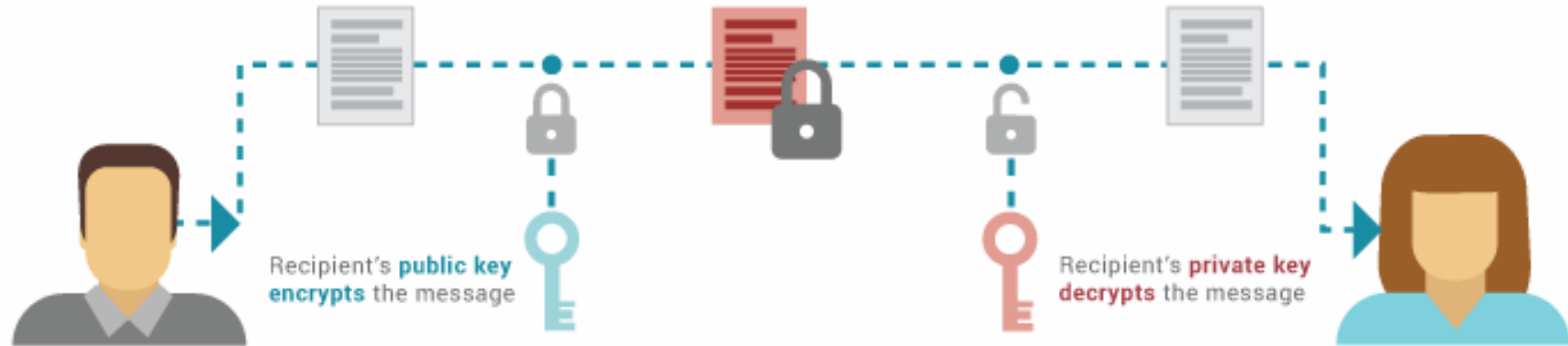


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Dat klopt voor jou!

<https://ssd.eff.org/en/module/key-concepts-encryption>

# PUBLIC KEY CRYPTOGRAPHY



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# Digital Signature and Certificate

Non-Repudiation



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# What is Digital Signatures?

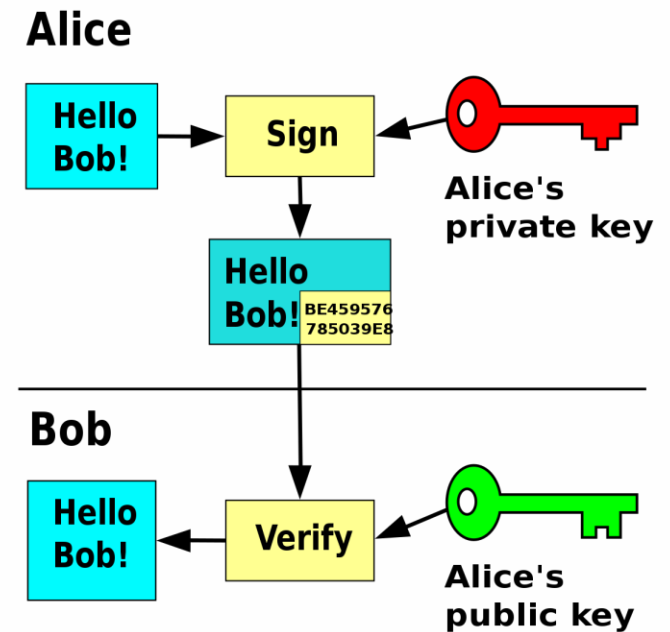
- Digital signature is a hash of a message that is encrypted with the sender's private key
- Digital signature, when done right can prove that:
  - A message was created by a known sender (sender authenticity),
  - And the message was not altered in transit (message integrity)
  - It is also used for non-repudiation



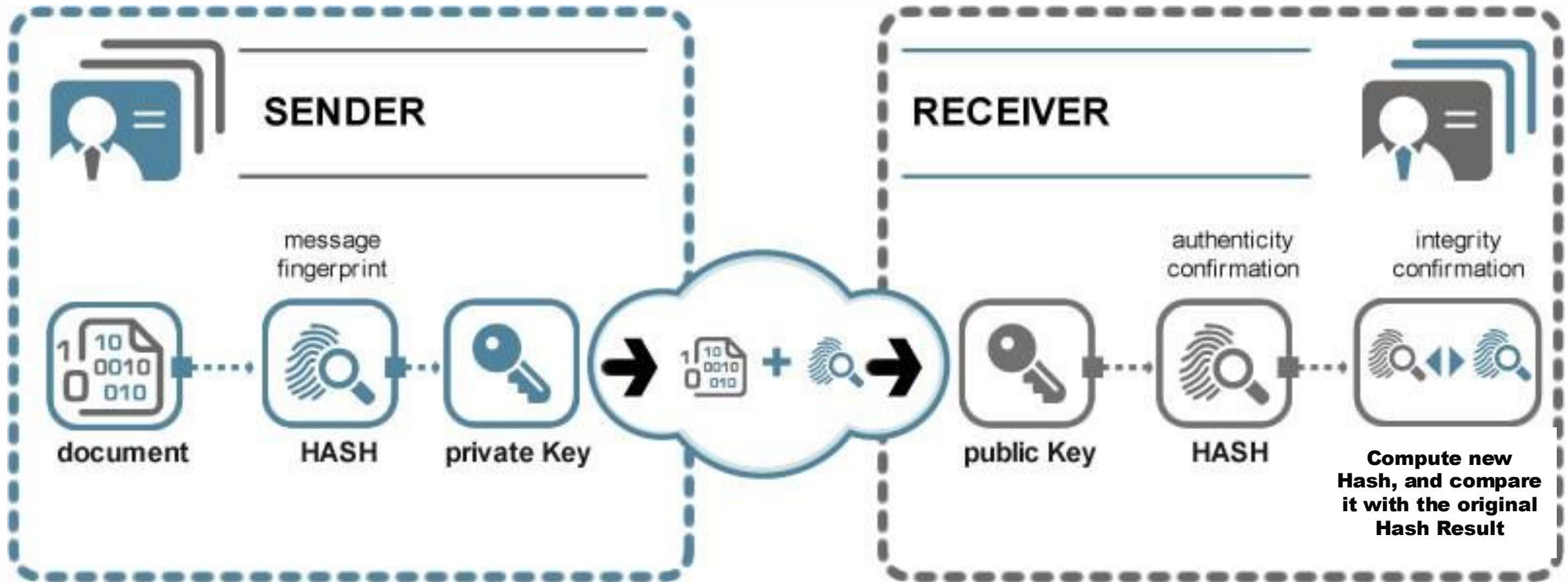


# How Digital Signatures Works

- After hashing the data, (to preserve integrity of the data) a hash algorithm uses public key algorithm to perform two further functions:
  - Sign the hash of the data ( to authenticate the sender) by encrypting the hash with the sender's private key
  - This prove the message was sent by the user with that private key and can be confirmed by using the Public key of the user to verify the hash



# To Verify Digital Signatures



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# Non Repudiation

Digital Signature

Digital Certificate



# How do we prove that this is the private, and public key pair of Mark Rutte



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public

private

BEGIN PUBLIC KEY BLOCK ----

```
-----BEGIN PUBLIC KEY BLOCK-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA
-----END PUBLIC KEY BLOCK-----
```

END PUBLIC KEY BLOCK ----

---- BEGIN PRIVATE KEY BLOCK

```
-----BEGIN PRIVATE KEY BLOCK-----
Proc-Type: 4, Private-Key-Type: 1
-----END PRIVATE KEY BLOCK-----
```

---- END PRIVATE KEY BLOCK



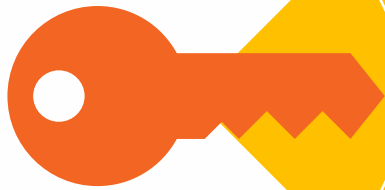
For non-repudiation we need to prove these key pair belongs to Mark Rutte

**Dat klopt voor jou!**

<https://ssd.eff.org/en/module/key-concepts-encryption>

---

# What is Digital Certificate?



It is an electronic document used to prove the ownership of a public key.



The document contains the following information:

- Information about the cryptography key
- The owner identity information ( The subject)
- The digital signature of the entity that verified the certificate's contents (the issuer).



# Certificate Authority (CA)

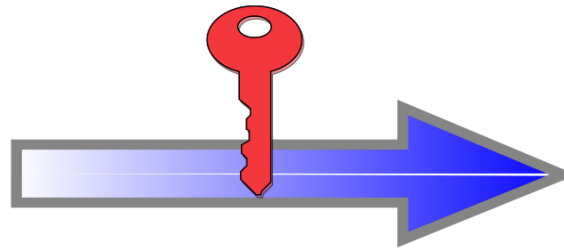
Identity Information and  
Public Key of Mark Rutte

|               |                  |
|---------------|------------------|
| Name:         | Mark Rutte       |
| Organization: | Dutch Government |
| Address:      | via .....        |
| Country:      | The Netherlands  |



Public Key  
of  
Mark Rutte

Certificate Authority  
verifies the identity of Mark Rutte  
and encrypts with its Private Key



Certificate of Mark Rutte

|               |                         |
|---------------|-------------------------|
| Name:         | Mark Rutte              |
| Organization: | Dutch Government        |
| Address:      | via .....               |
| Country:      | The Netherlands         |
| Validity:     | 1997/07/01 - 2047/06/30 |



Public Key  
of  
Mark Rutte

Digital Signature  
of the Certificate Authority

Digitally Signed by  
Certificate Authority

→ A CA is a trusted third-party responsible for the issue, maintenance, and validation of identity of entities, persons, websites or companies

→ The CA binds these information to the cryptographic keys (public and Private Keys) through the issuance of electronic documents known as digital certificates



---

# Email Security





# Email Encryption standards (PGP)

- Pretty Good Privacy (PGP)
  - Combines the CA hierarchy with the “web of trust” concept
  - One should become trusted by one or more PGP users to begin using the system
  - Available in two versions
    - Commercial version:
      - Uses RSA for key exchange, IDEA for encryption/decryption, MD5 for HASH
    - Open-source version:
      - Uses DH for key exchange, CAS for encryption/decryption, SHA-1 for HASH algorithm





---

# PGP Limitations



The sender and receiver information is unencrypted

Encrypting the sender and receiver information would break email

PGP does not encrypt the subject line of your emails



---

# Email Encryption standards S/MIME

- Secure/Multipurpose Internet Mail Extensions
- Public encryption and signing of MIME data such as email messages to verify a sender's identity
- S/MIME
  - Defacto standard for encrypted email
  - Uses RSA encryption
  - Relies on use of X.509 certificates for exchanging keys
  - Supports AES and 3DES symmetric algorithm



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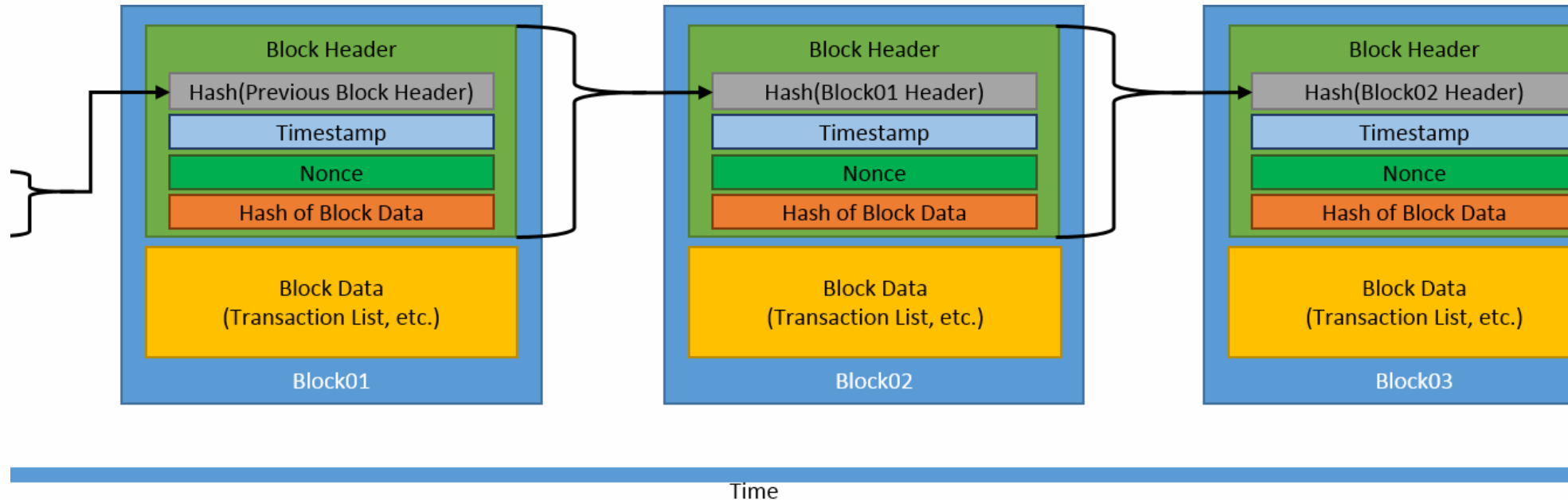
# Blockchain



- 
- A blockchain is a collaborative, tamper-resistant ledger that maintains transactional records.
  - The transactional records (data) are grouped into blocks
  - A block is connected to the previous one by including a unique identifier that is based on the previous block's data.
  - As a result, if the data is changed in one block, it's unique identifier changes, which can be seen in every subsequent block (providing tamper evidence).
  - This domino effect allows all users within the blockchain to know if a previous block's data has been tampered with.
  - Since a blockchain network is difficult to alter or destroy, it provides a resilient method of collaborative record keeping



# How The Blockchain Uses Cryptography





Hash: 6U9P2

Hash: 8Y5C9

Hash: 9L4Z1



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# Exercise 3

What to use...?



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# Other Cryptography Topics To Learn

- Database Privacy and Security
- Cryptoanalyses
- Computing over encrypted data
- Universally Composable Security
- Homomorphic encryption
- Quantum Cryptography
- Many more.....







---

# Vijfhart

Rokus Janssen, adviseur en accountmanager bij Vijfhart voor KPN



Heb je vragen of opmerkingen, neem dan vooral contact op met mij voor passend advies. Je kan mij bereiken via:

E: [r.janssen@5hart.nl](mailto:r.janssen@5hart.nl)

T: 088 542 78 88

Graag tot snel!



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**Dat klopt voor jou!**